

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

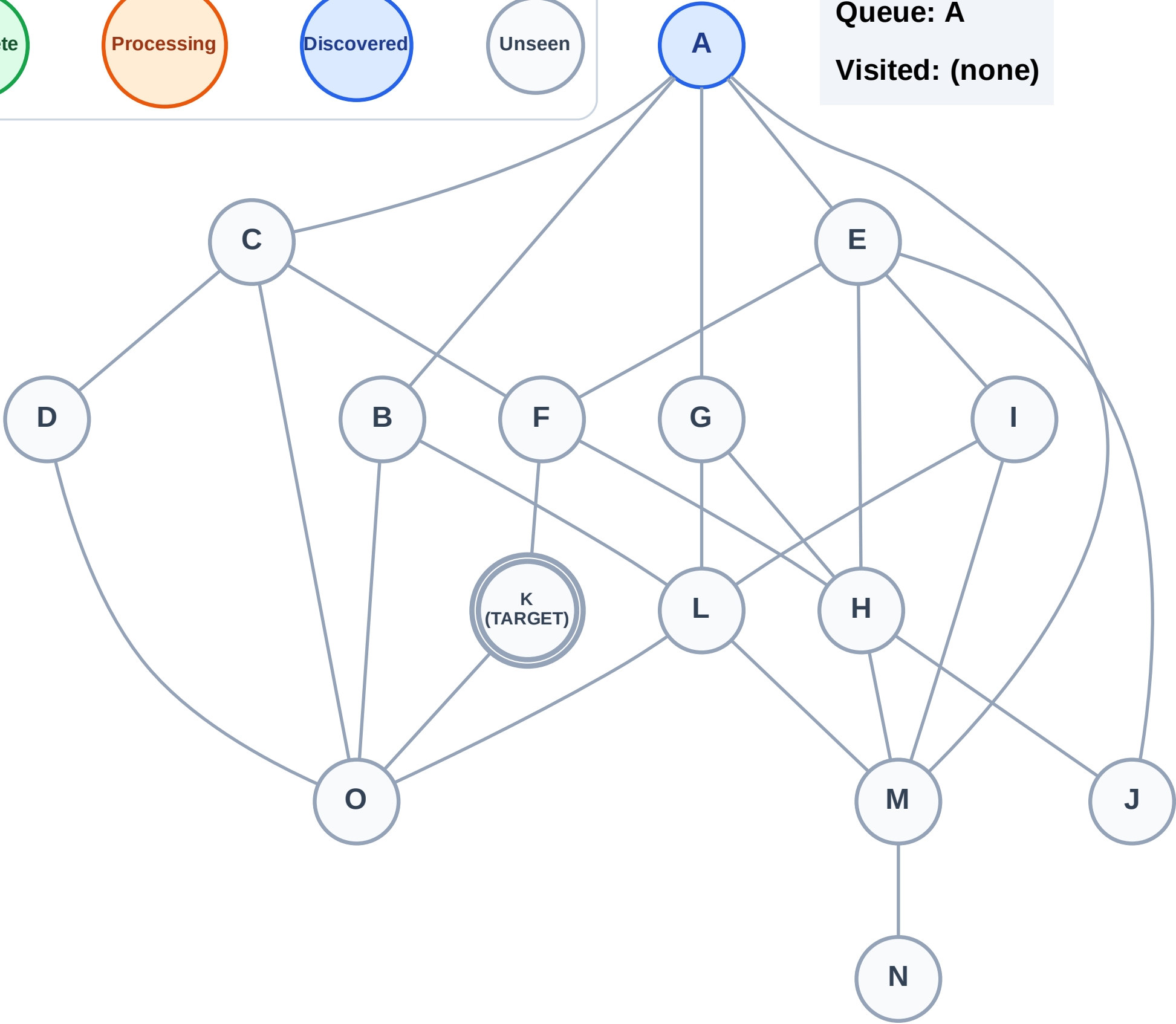
Complete

Processing

Discovered

Unseen

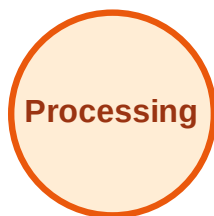
Queue: A
Visited: (none)



BFS Traversal

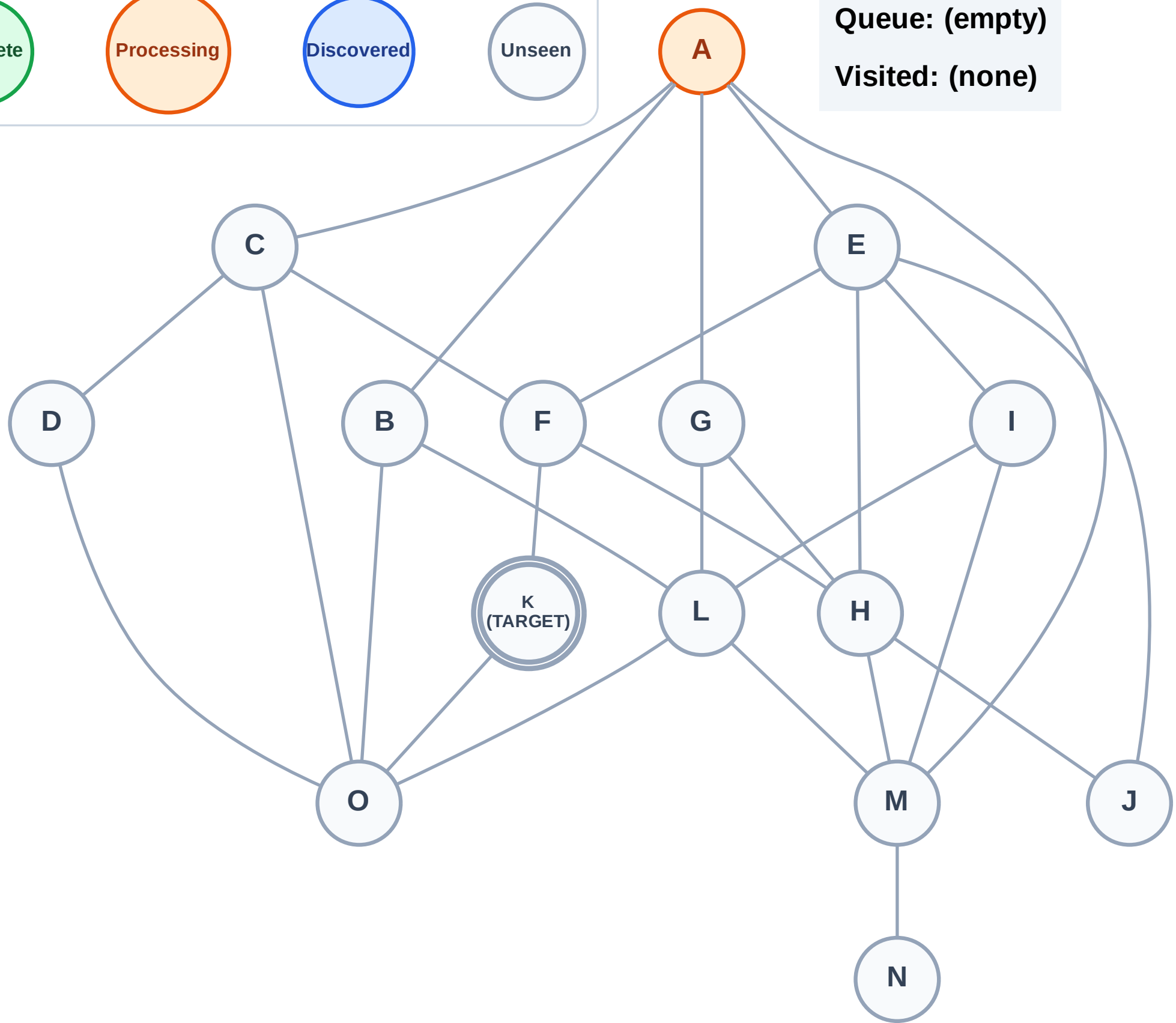
Step 2: Process node A

Legend



Queue: (empty)

Visited: (none)



BFS Traversal

Step 3: Discover B, C, E, G, M from A

Legend

Complete

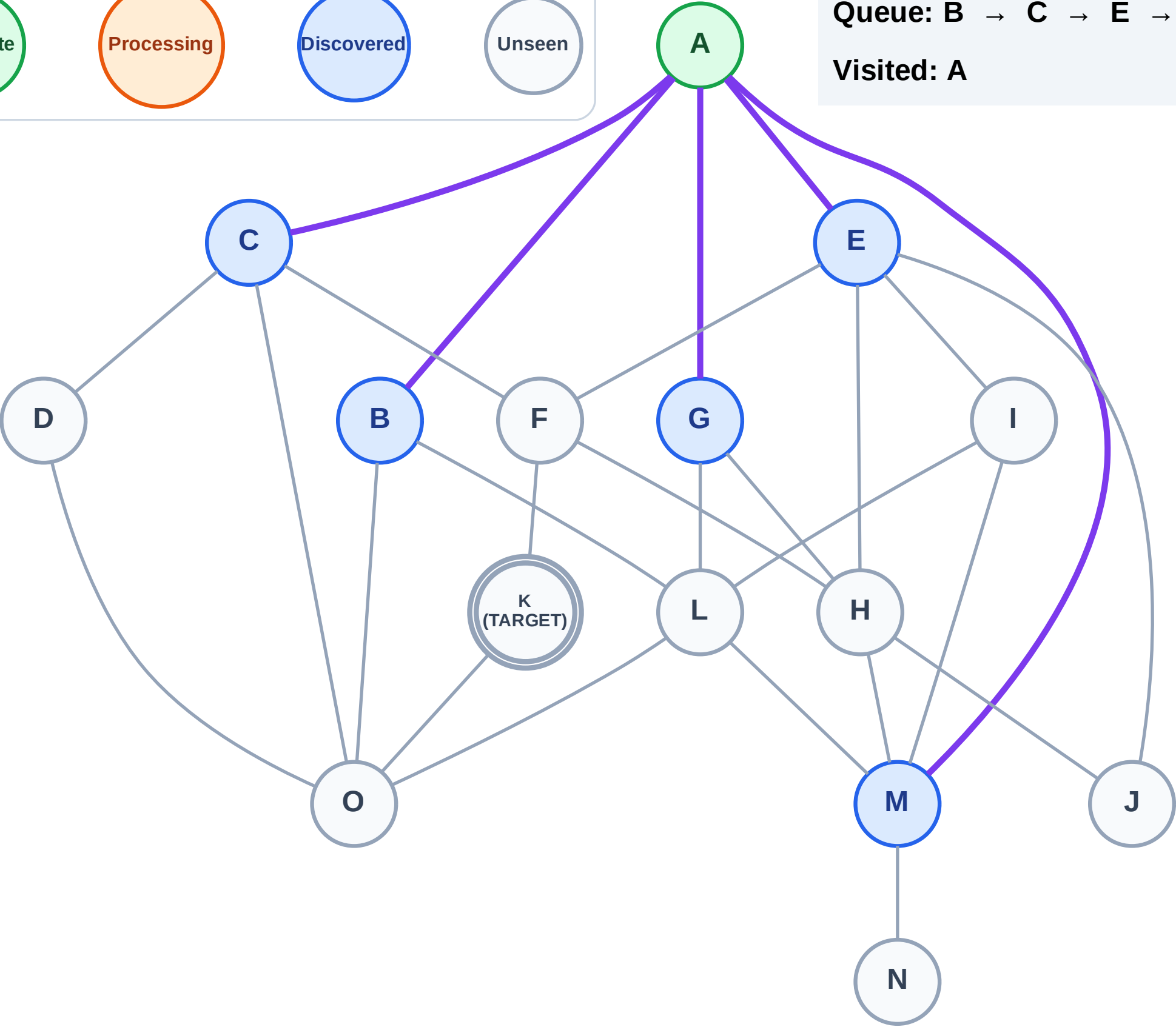
Processing

Discovered

Unseen

Queue: B → C → E → G → M

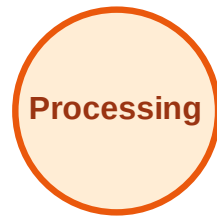
Visited: A



BFS Traversal

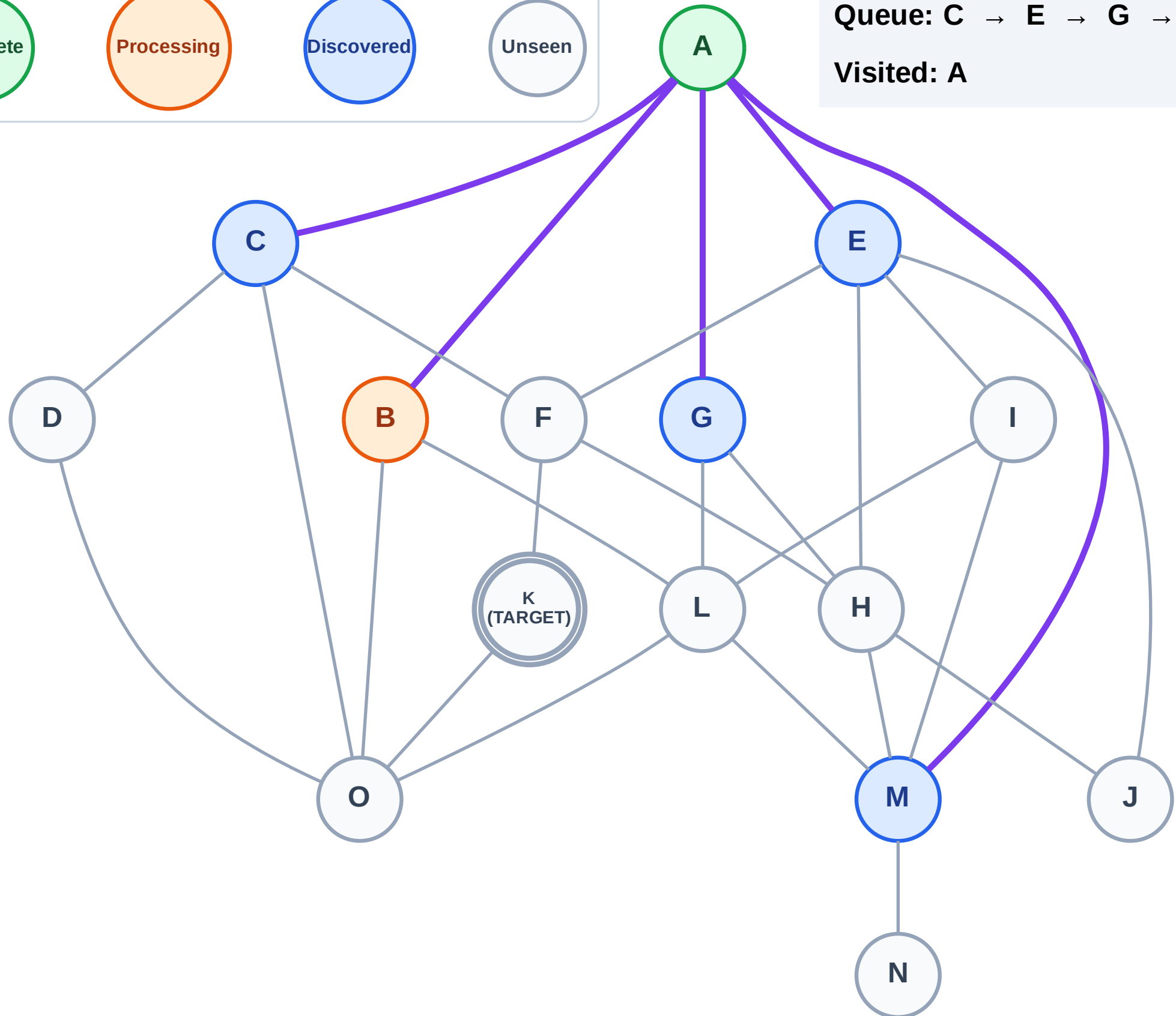
Step 4: Process node B

Legend



Queue: C → E → G → M

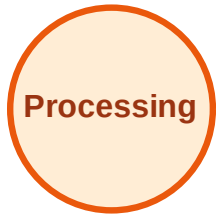
Visited: A



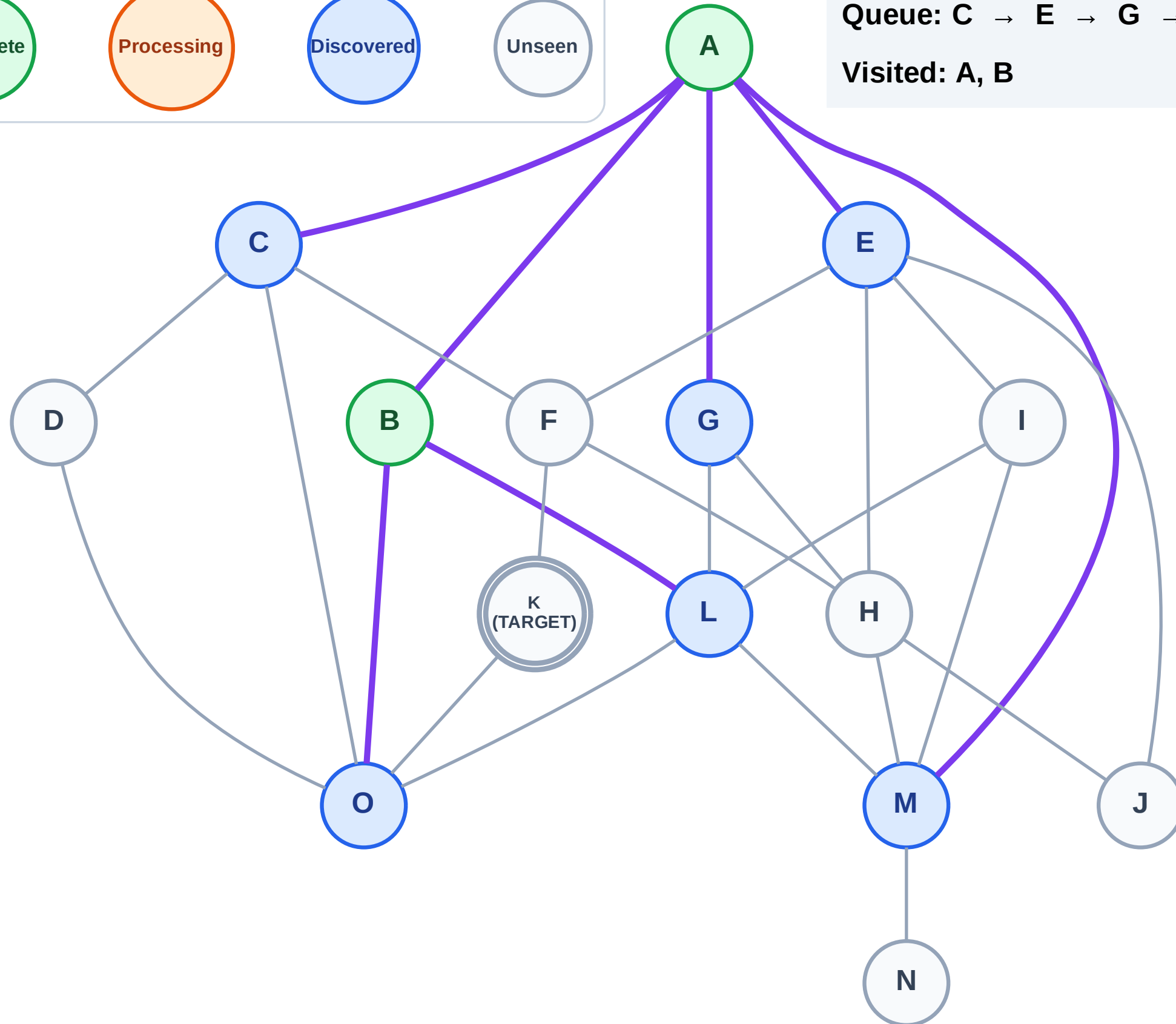
BFS Traversal

Step 5: Discover L, O from B

Legend



Queue: C → E → G → M → L → O
Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

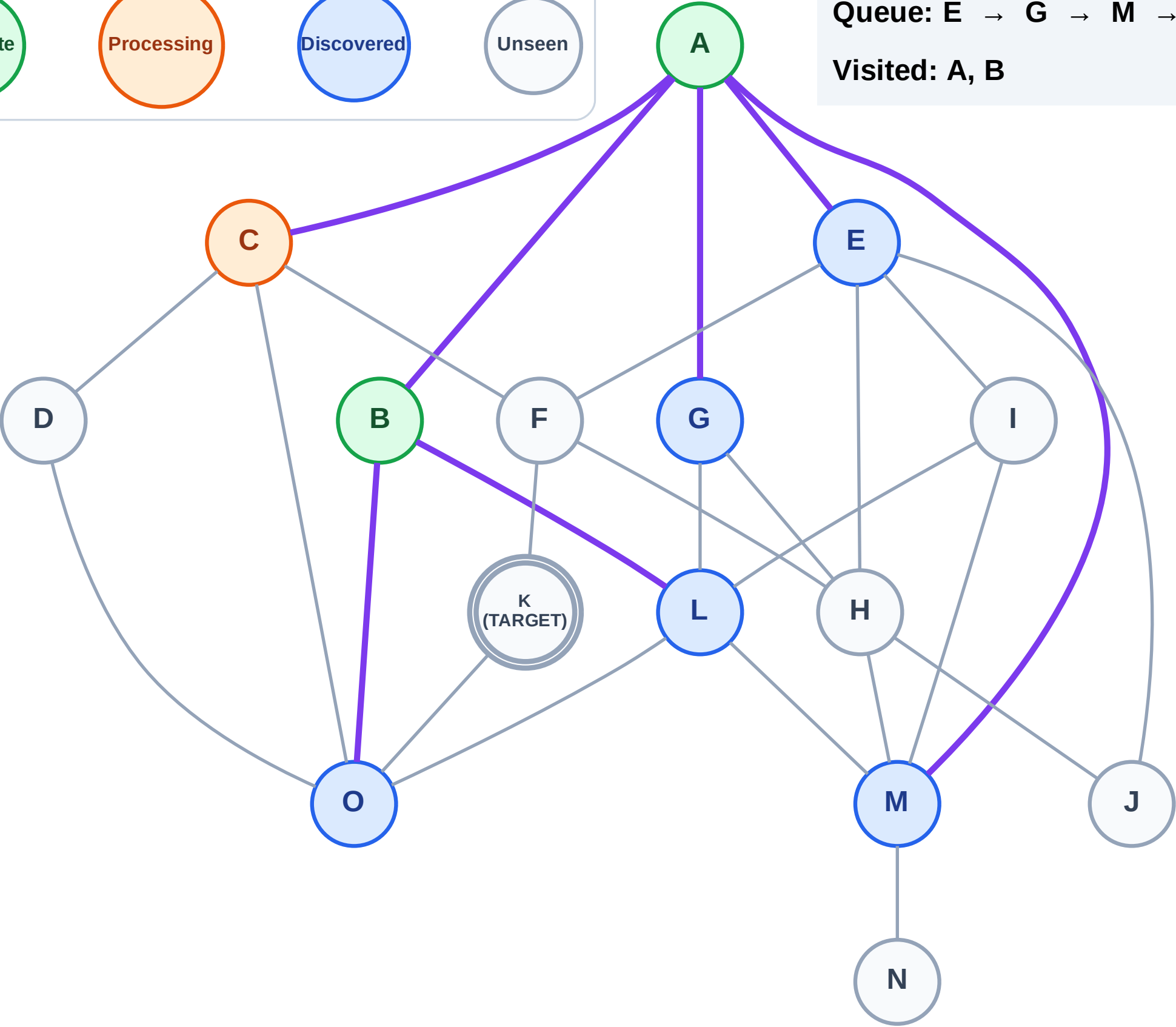
Processing

Discovered

Unseen

Queue: E → G → M → L → O

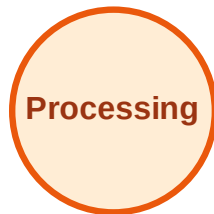
Visited: A, B



BFS Traversal

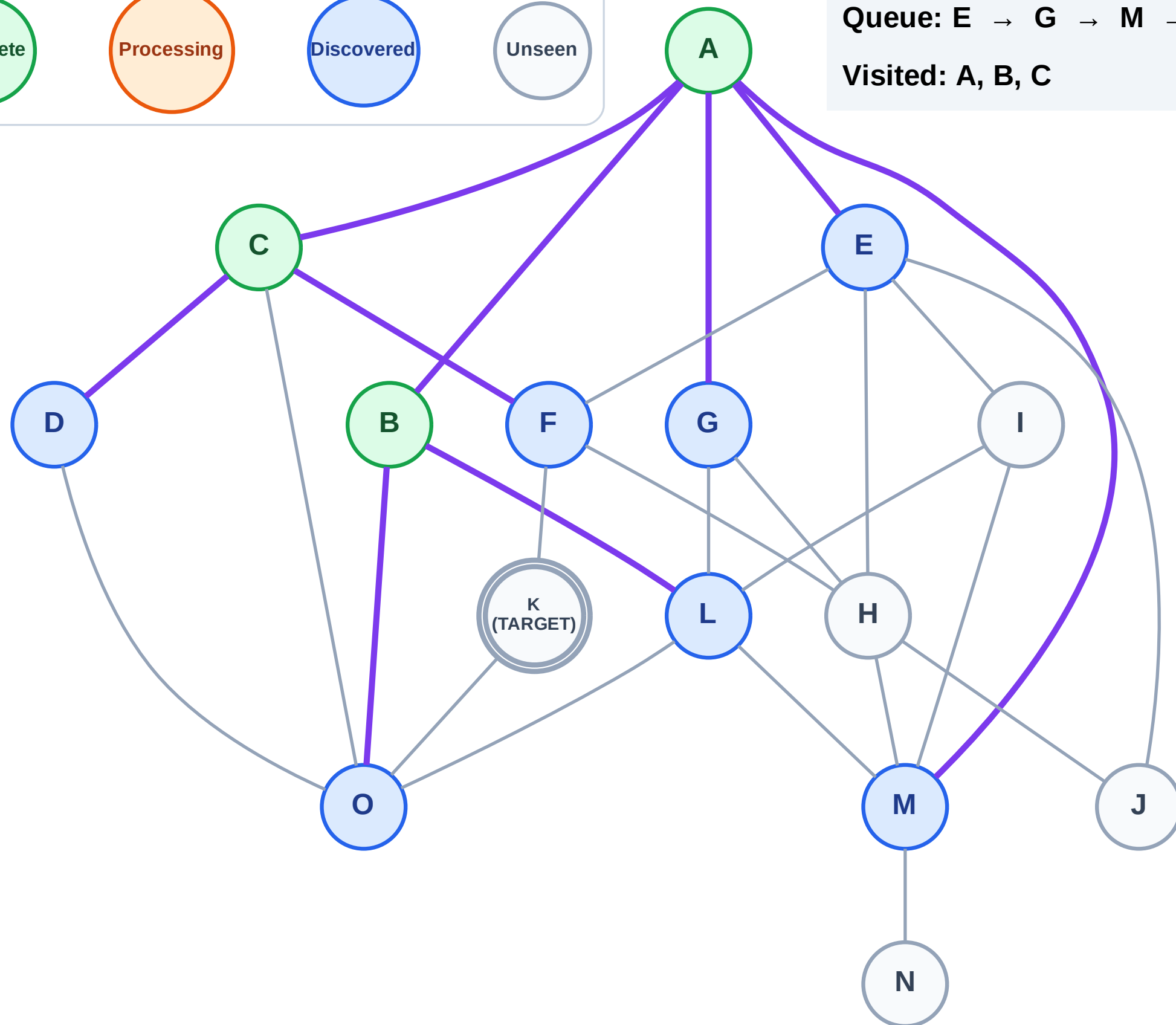
Step 7: Discover D, F from C

Legend



Queue: E → G → M → L → O → D → F

Visited: A, B, C



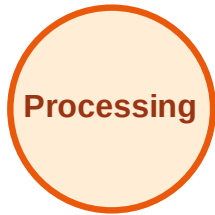
BFS Traversal

Step 8: Process node E

Legend



Complete



Processing



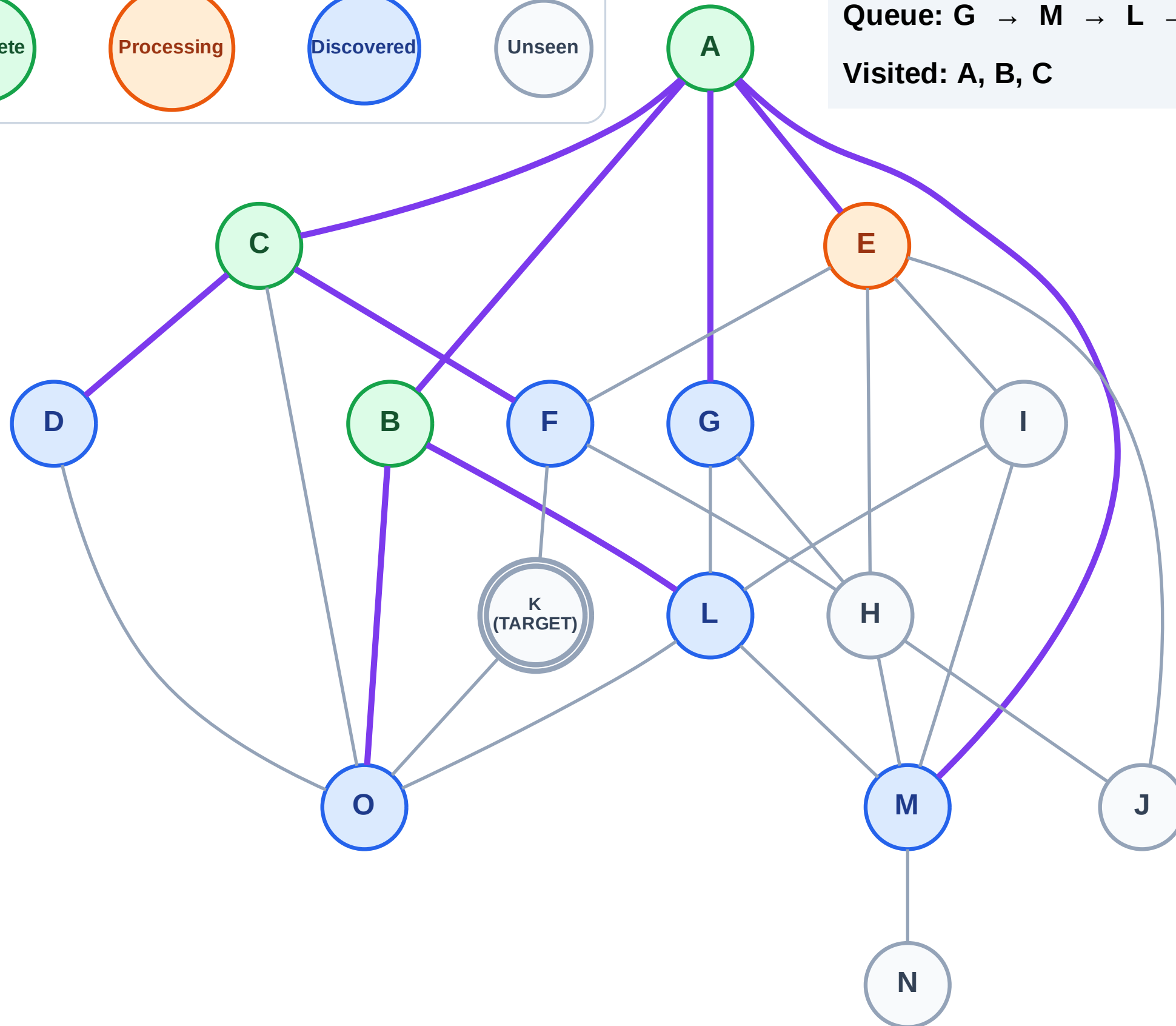
Discovered



Unseen

Queue: G → M → L → O → D → F

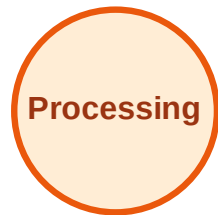
Visited: A, B, C



BFS Traversal

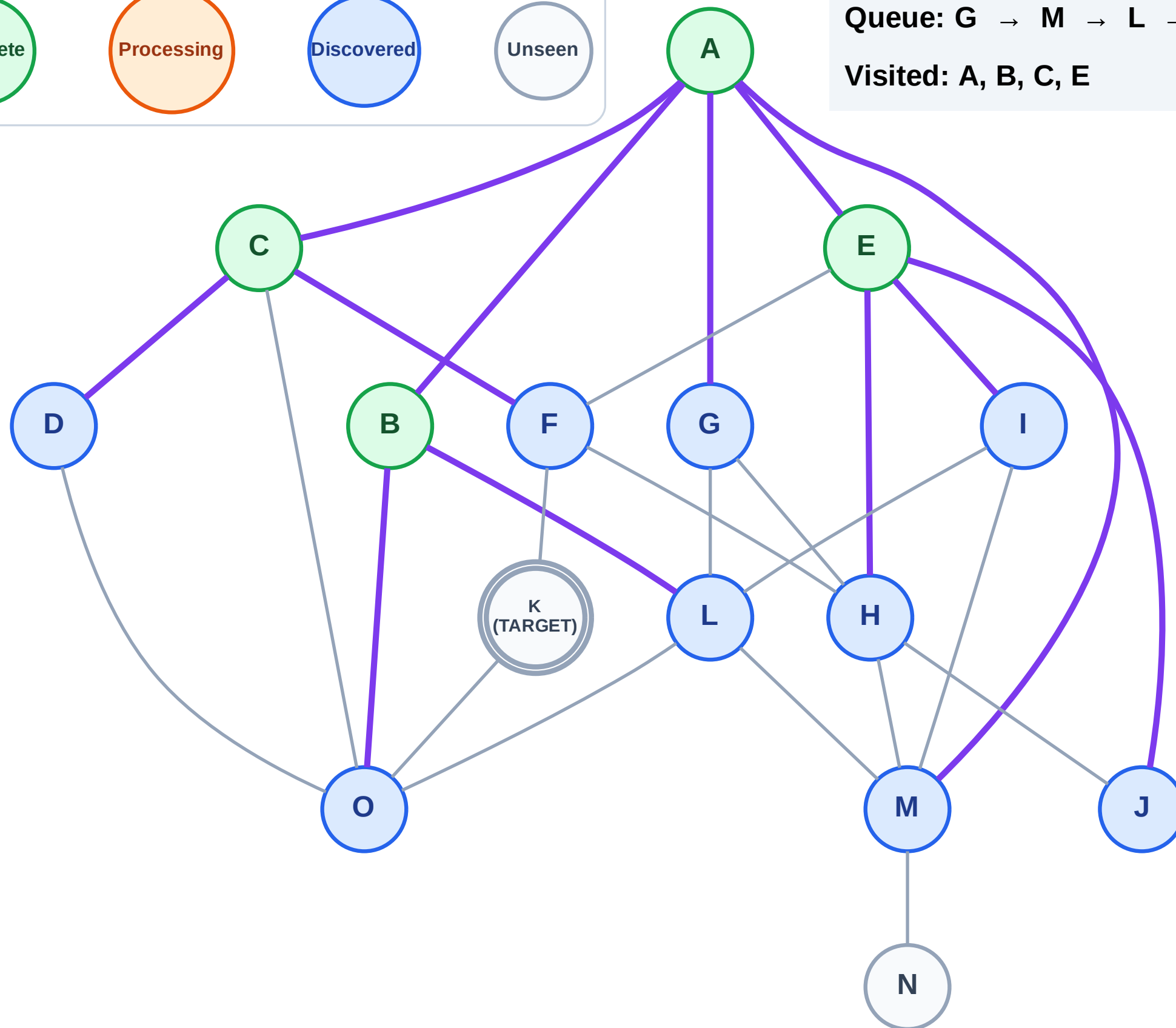
Step 9: Discover H, I, J from E

Legend



Queue: G → M → L → O → D → F → H → I → J

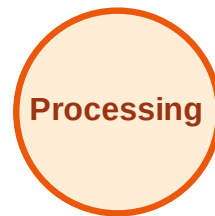
Visited: A, B, C, E



BFS Traversal

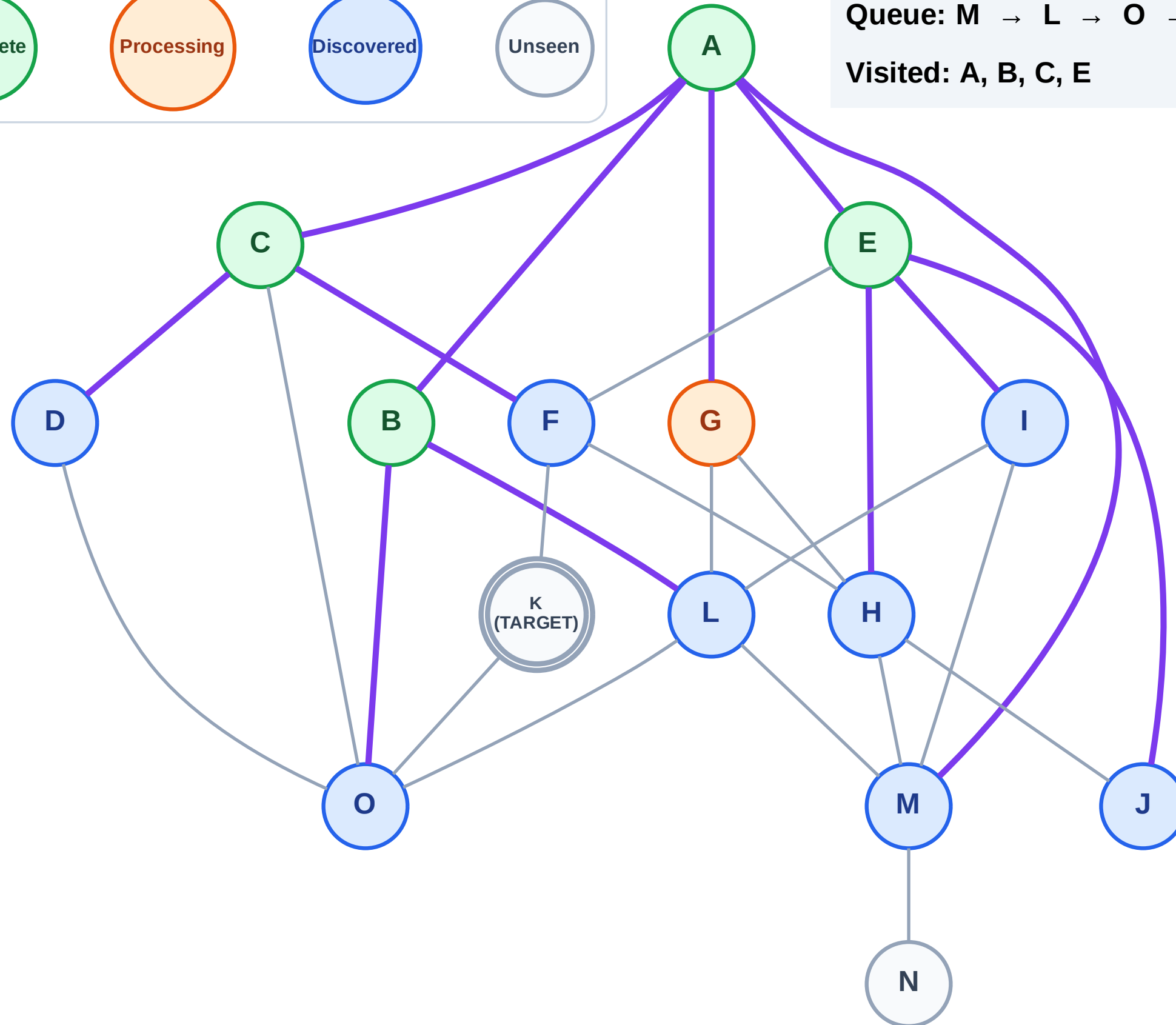
Step 10: Process node G

Legend



Queue: M → L → O → D → F → H → I → J

Visited: A, B, C, E



BFS Traversal

Step 11: No new neighbors from G

Legend

Complete

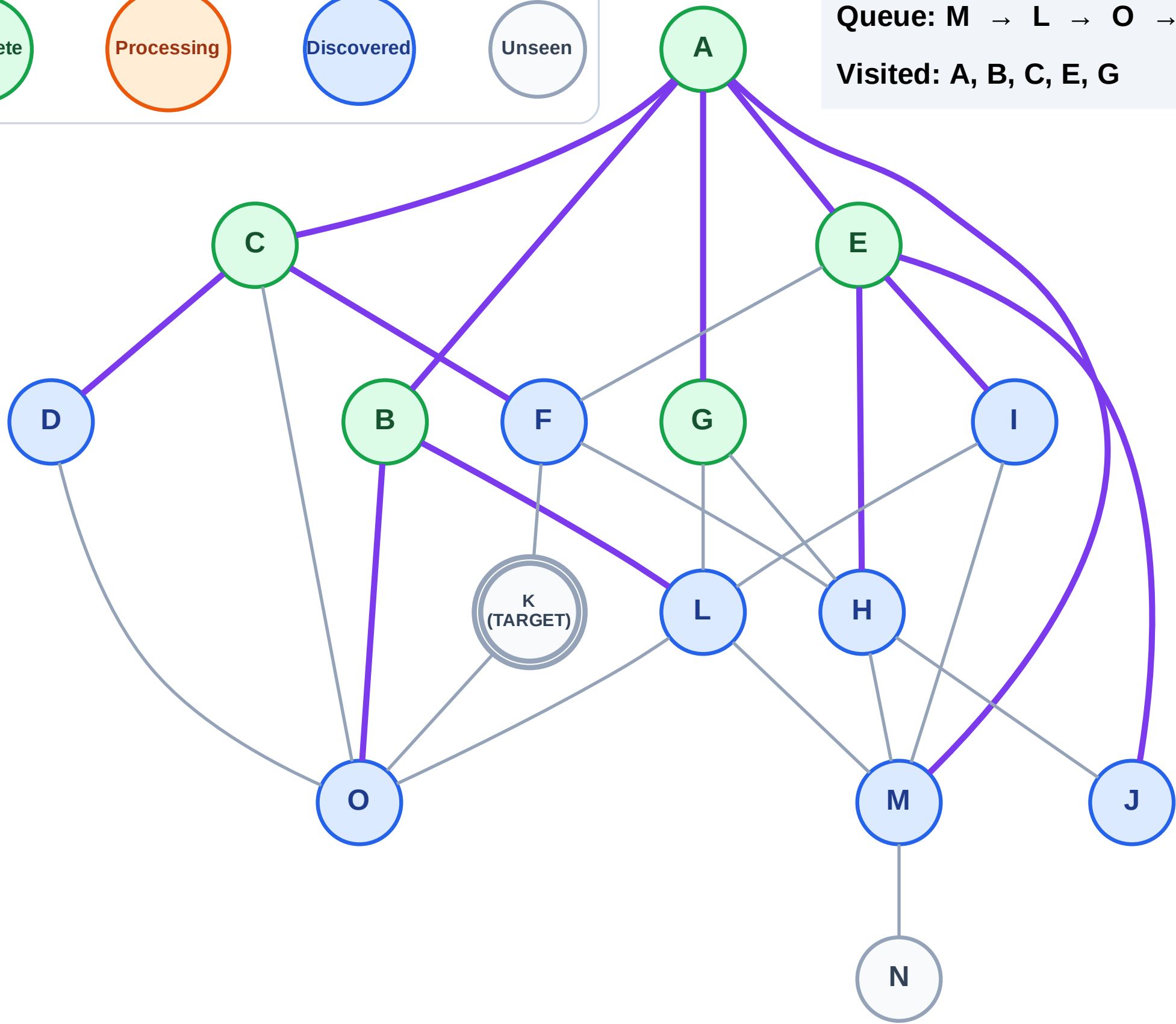
Processing

Discovered

Unseen

Queue: M → L → O → D → F → H → I → J

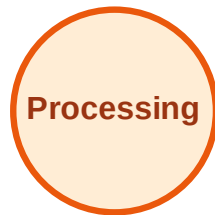
Visited: A, B, C, E, G



BFS Traversal

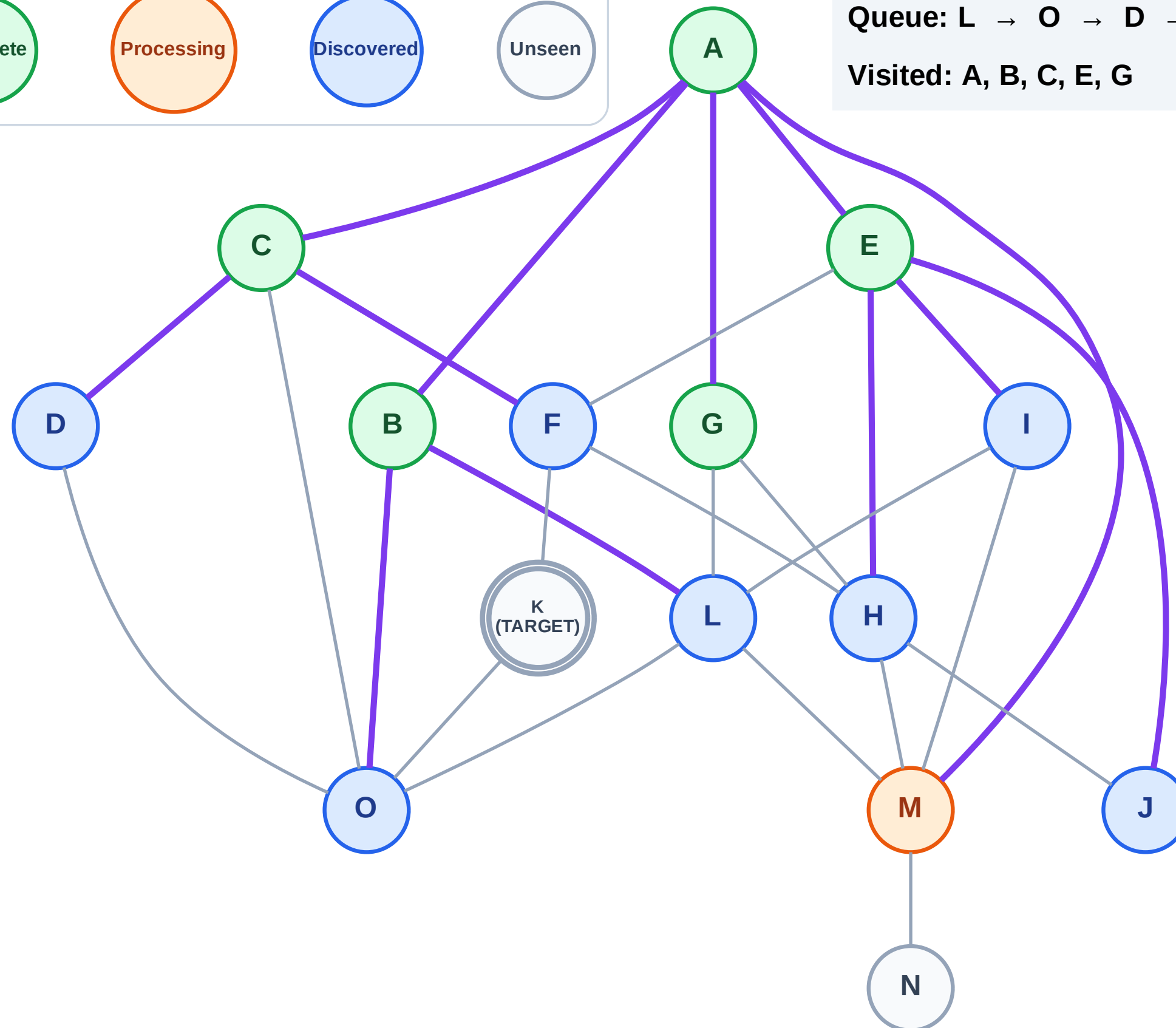
Step 12: Process node M

Legend



Queue: L → O → D → F → H → I → J

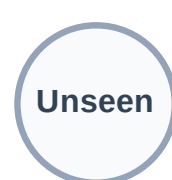
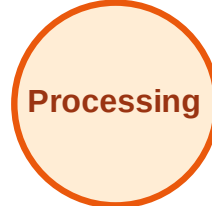
Visited: A, B, C, E, G



BFS Traversal

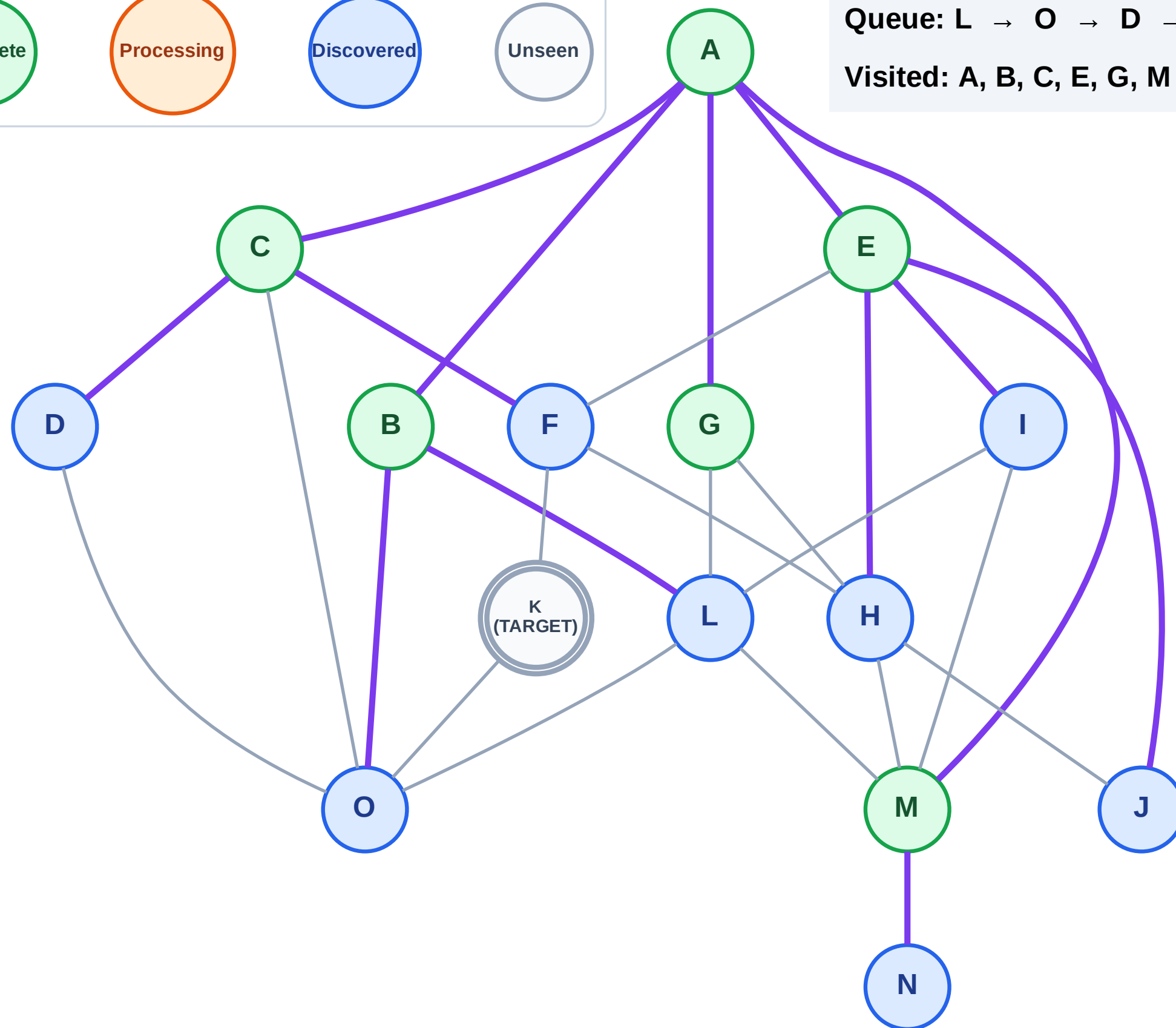
Step 13: Discover N from M

Legend



Queue: L → O → D → F → H → I → J → N

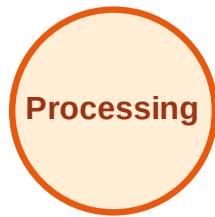
Visited: A, B, C, E, G, M



BFS Traversal

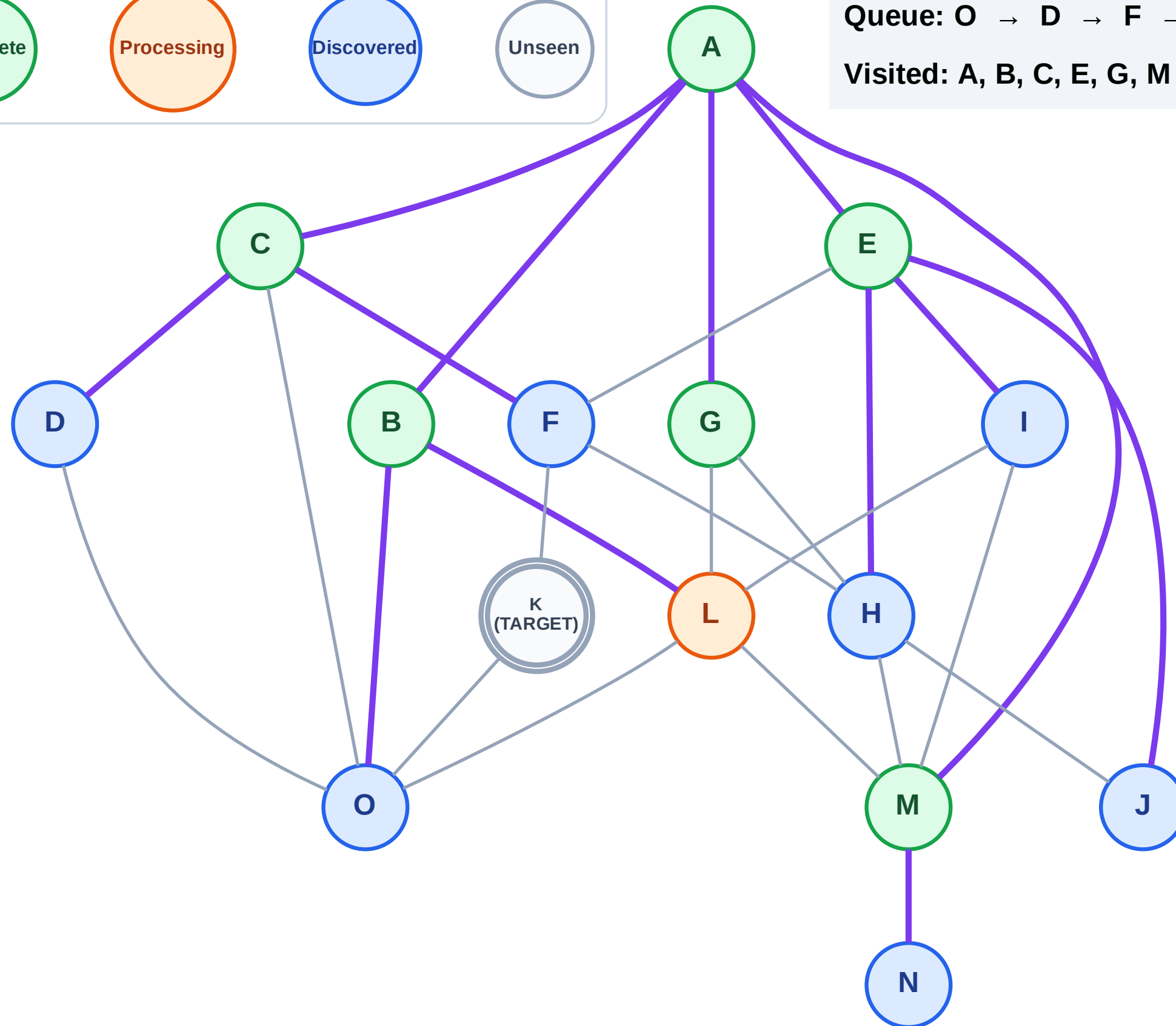
Step 14: Process node L

Legend



Queue: O → D → F → H → I → J → N

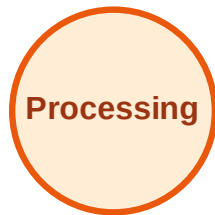
Visited: A, B, C, E, G, M



BFS Traversal

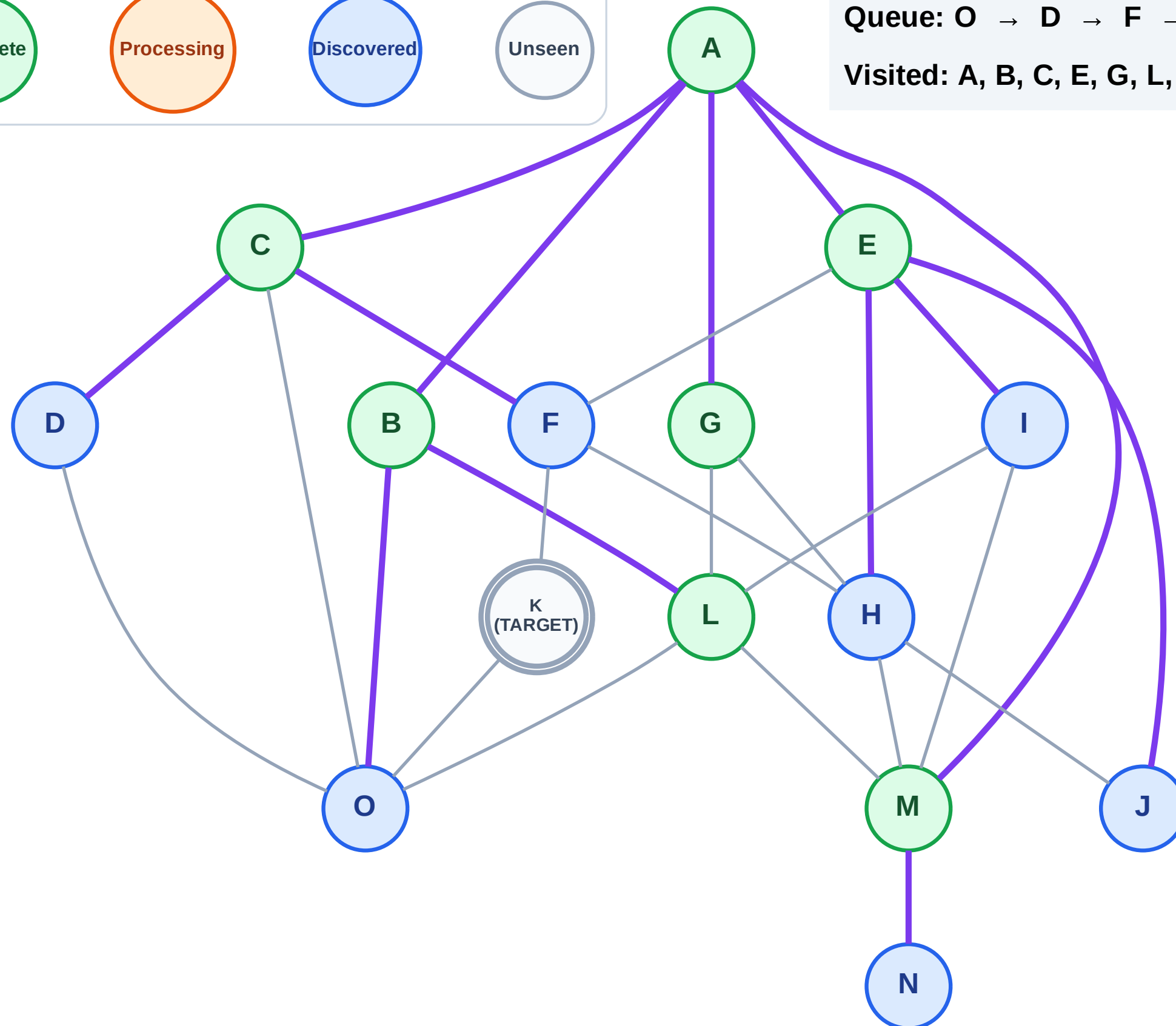
Step 15: No new neighbors from L

Legend



Queue: O → D → F → H → I → J → N

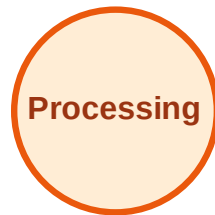
Visited: A, B, C, E, G, L, M



BFS Traversal

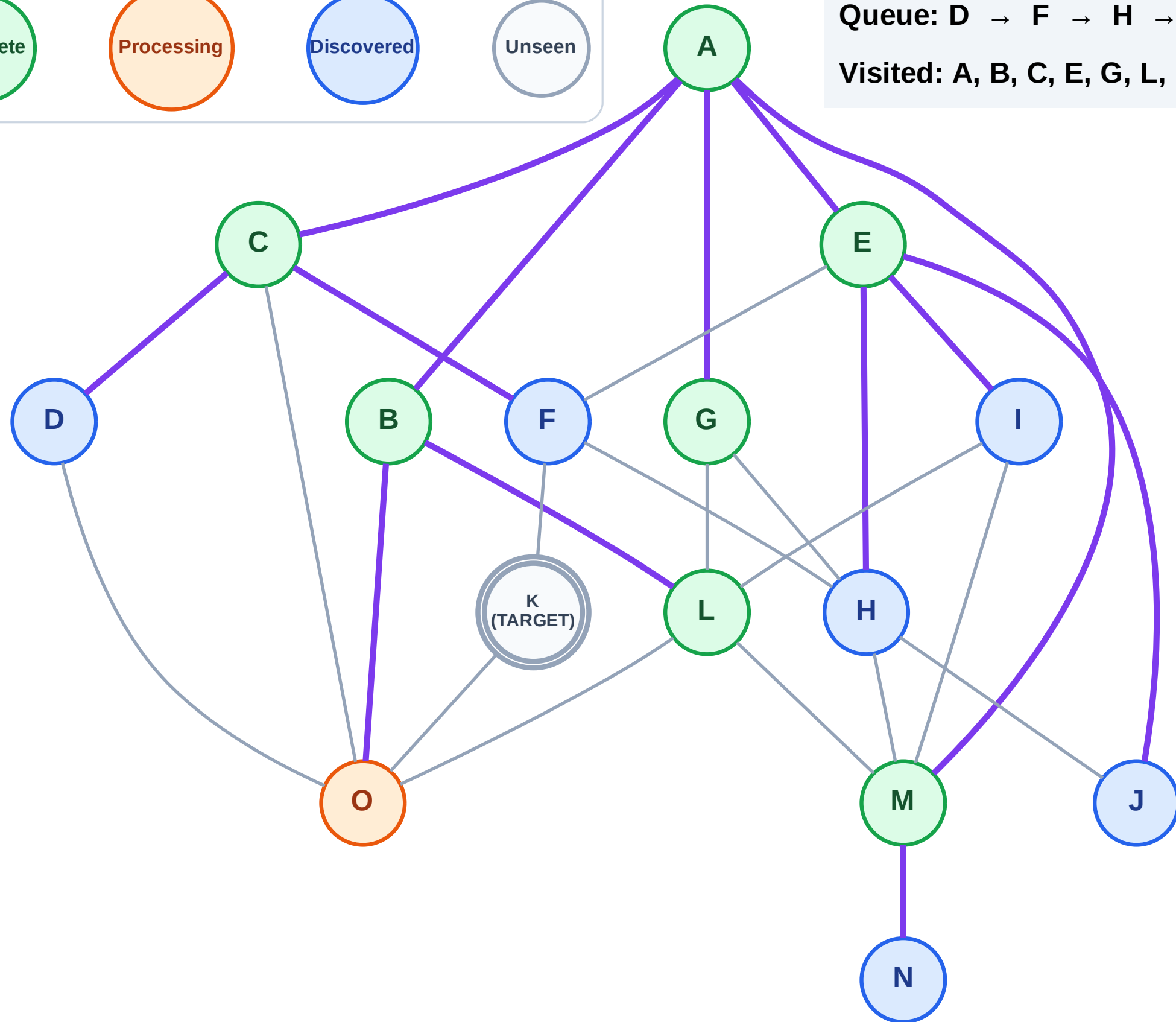
Step 16: Process node O

Legend



Queue: D → F → H → I → J → N

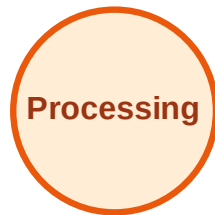
Visited: A, B, C, E, G, L, M



BFS Traversal

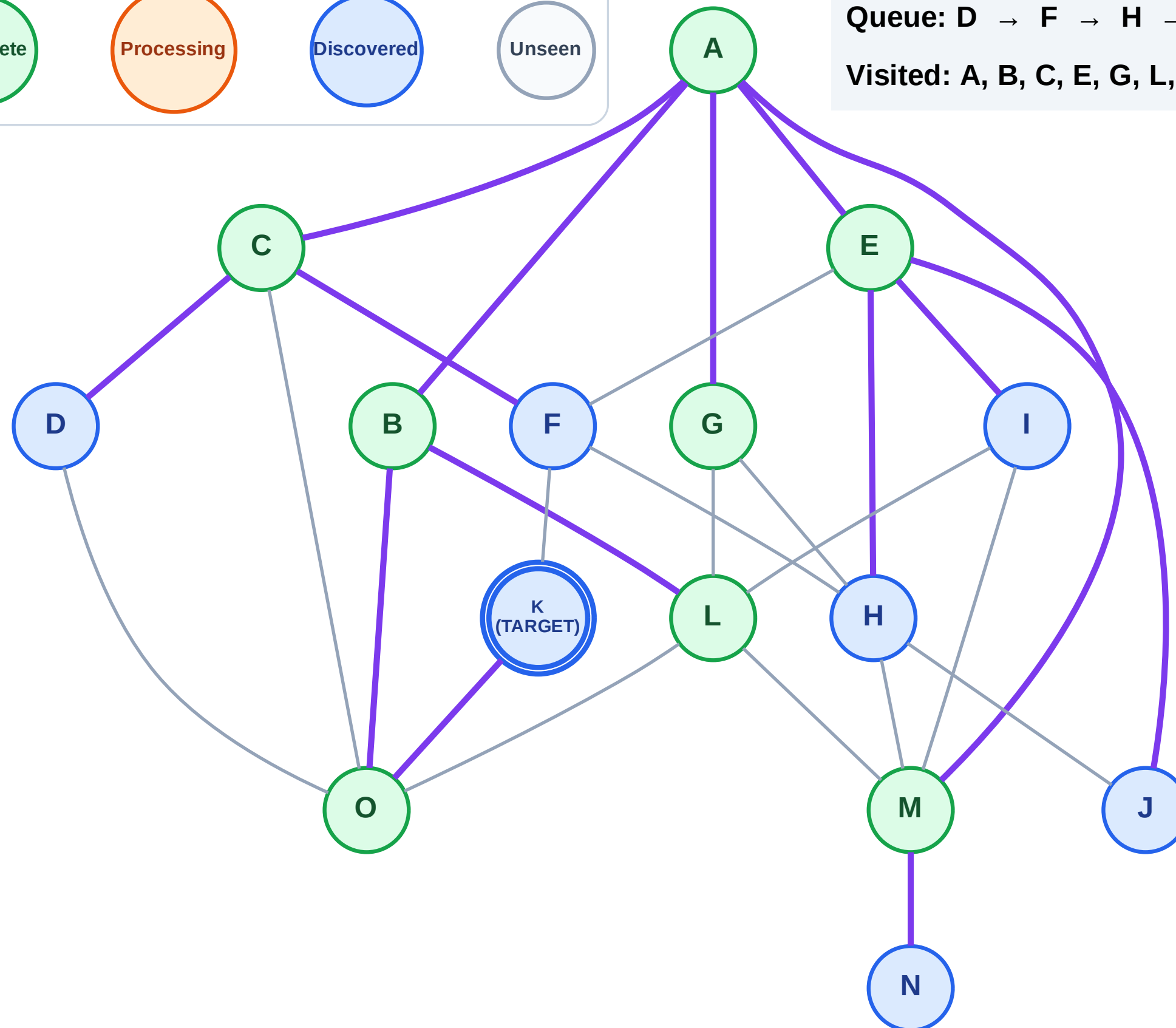
Step 17: Discover K from O

Legend



Queue: D → F → H → I → J → N → K

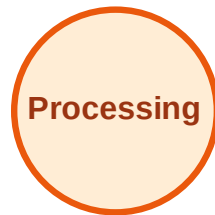
Visited: A, B, C, E, G, L, M, O



BFS Traversal

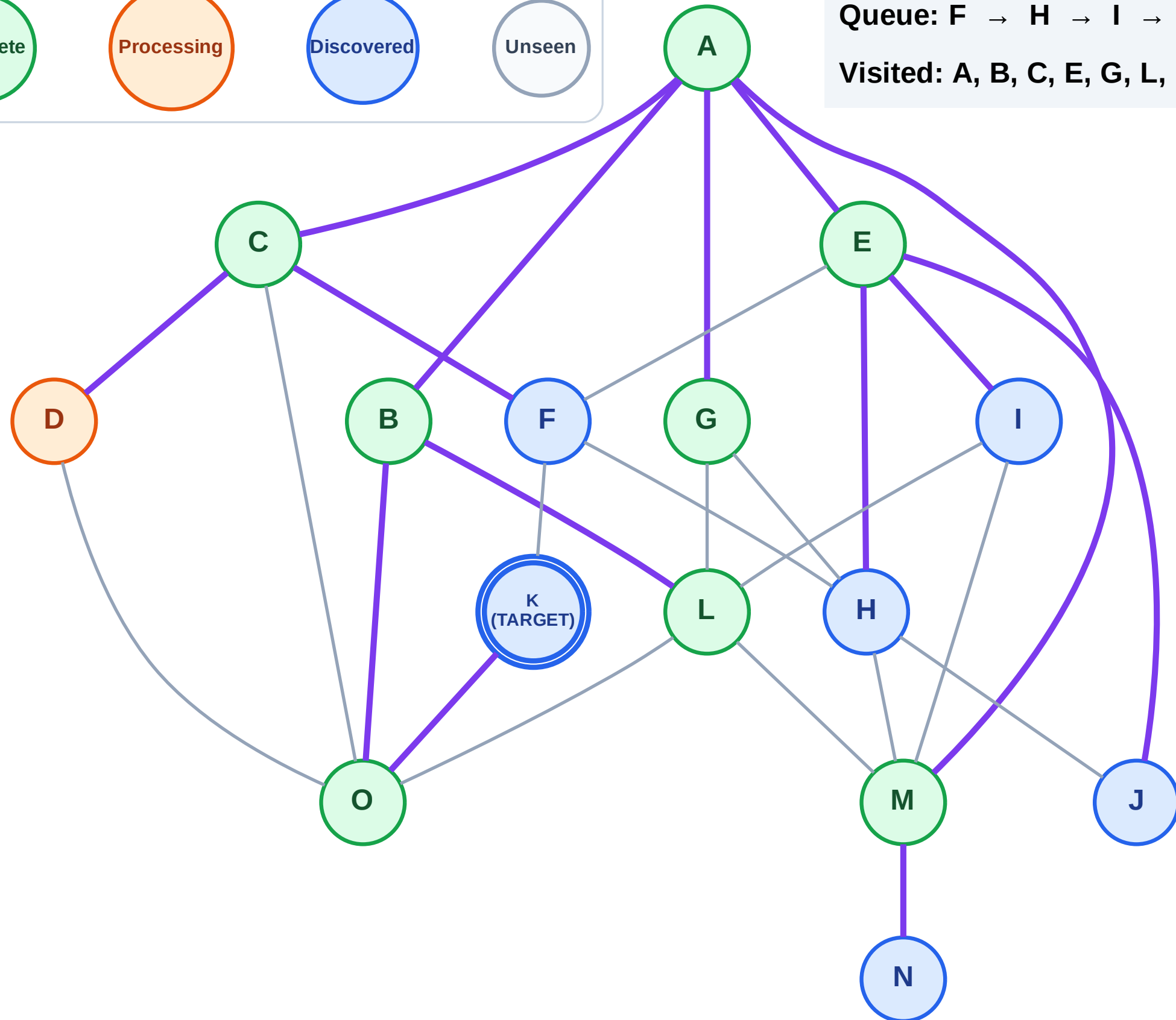
Step 18: Process node D

Legend



Queue: F → H → I → J → N → K

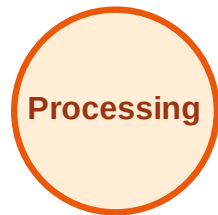
Visited: A, B, C, E, G, L, M, O



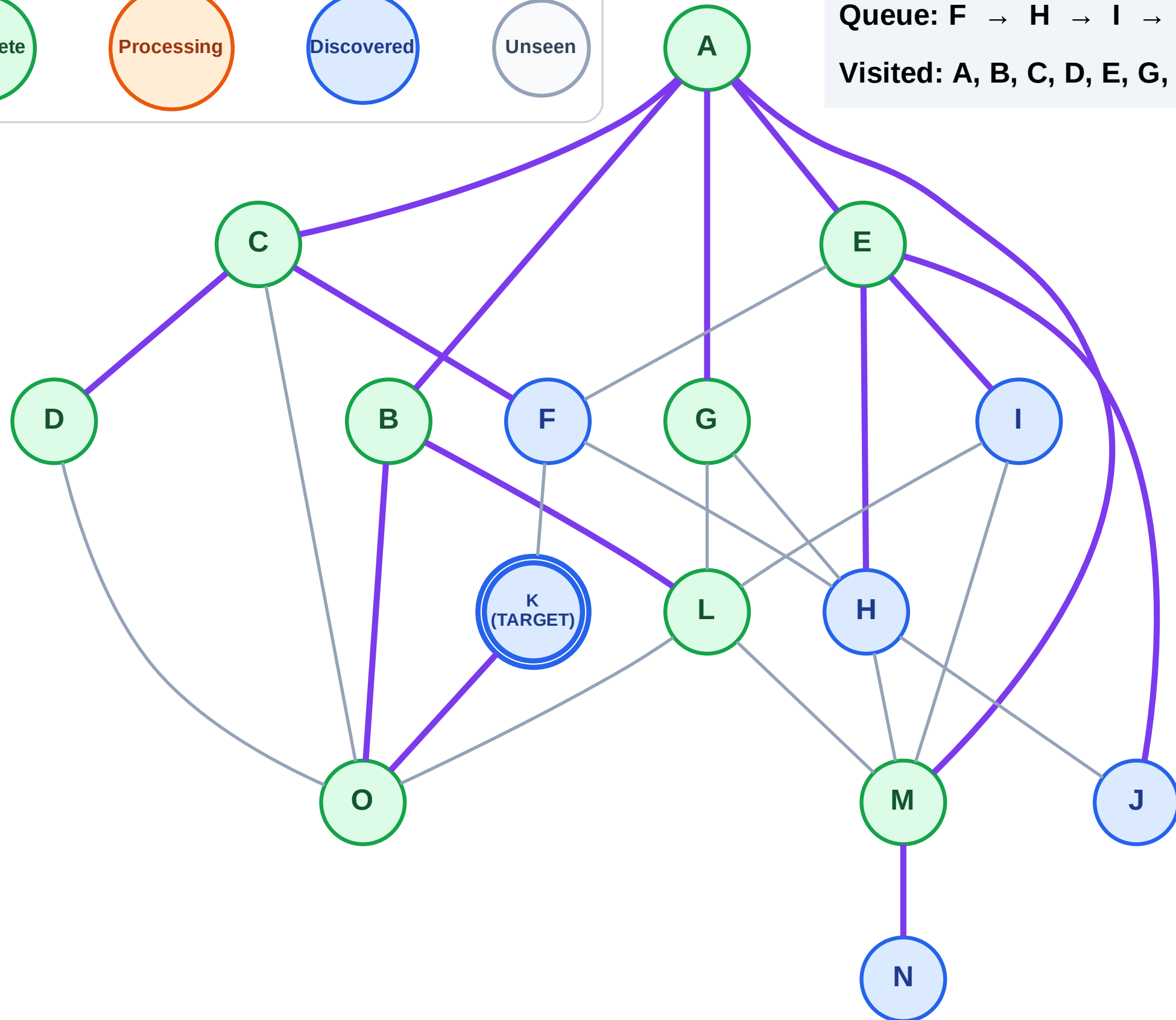
BFS Traversal

Step 19: No new neighbors from D

Legend



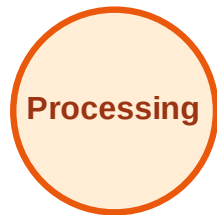
Queue: F → H → I → J → N → K
Visited: A, B, C, D, E, G, L, M, O



BFS Traversal

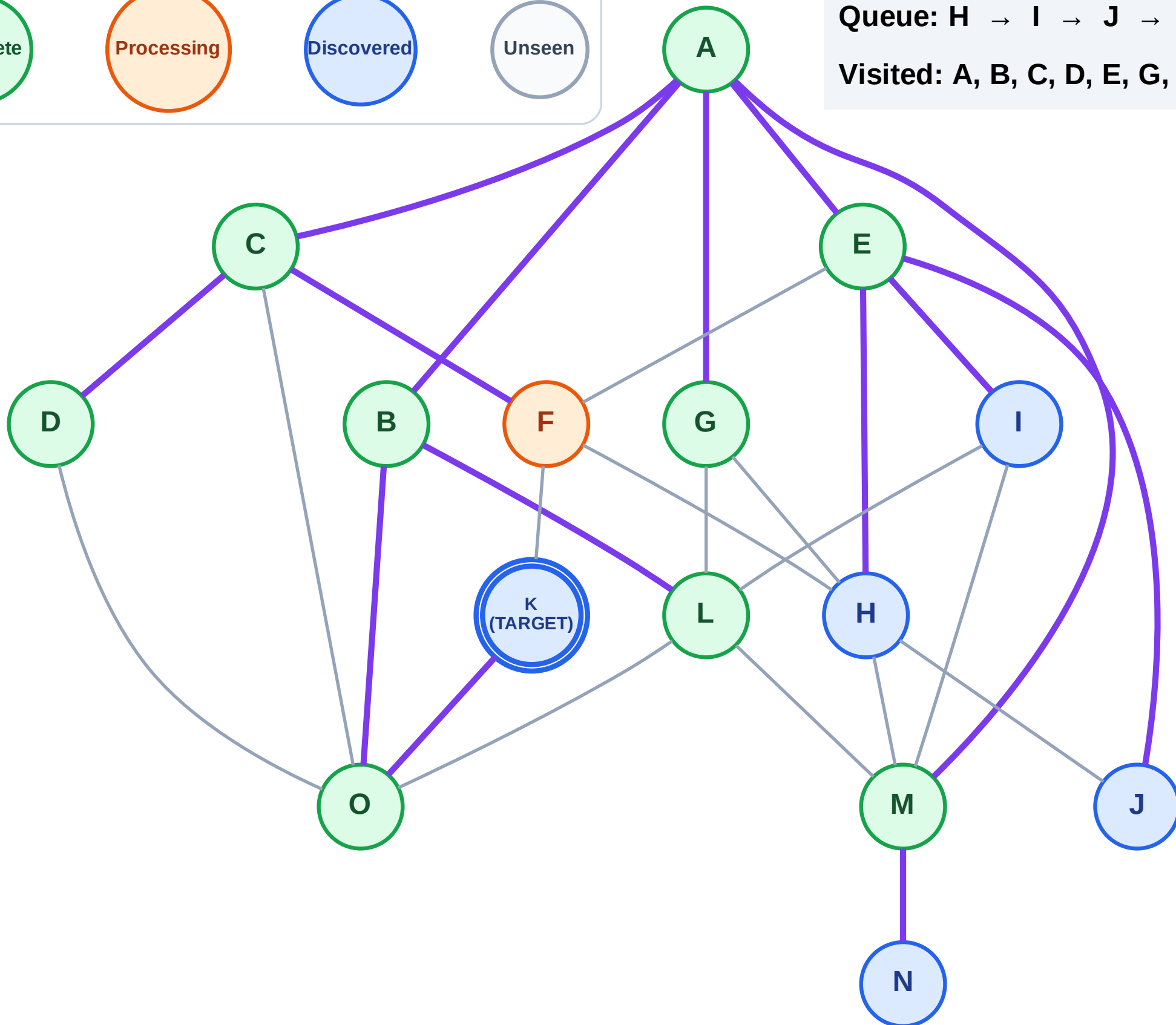
Step 20: Process node F

Legend



Queue: H → I → J → N → K

Visited: A, B, C, D, E, G, L, M, O



BFS Traversal

Step 21: No new neighbors from F

Legend

Complete

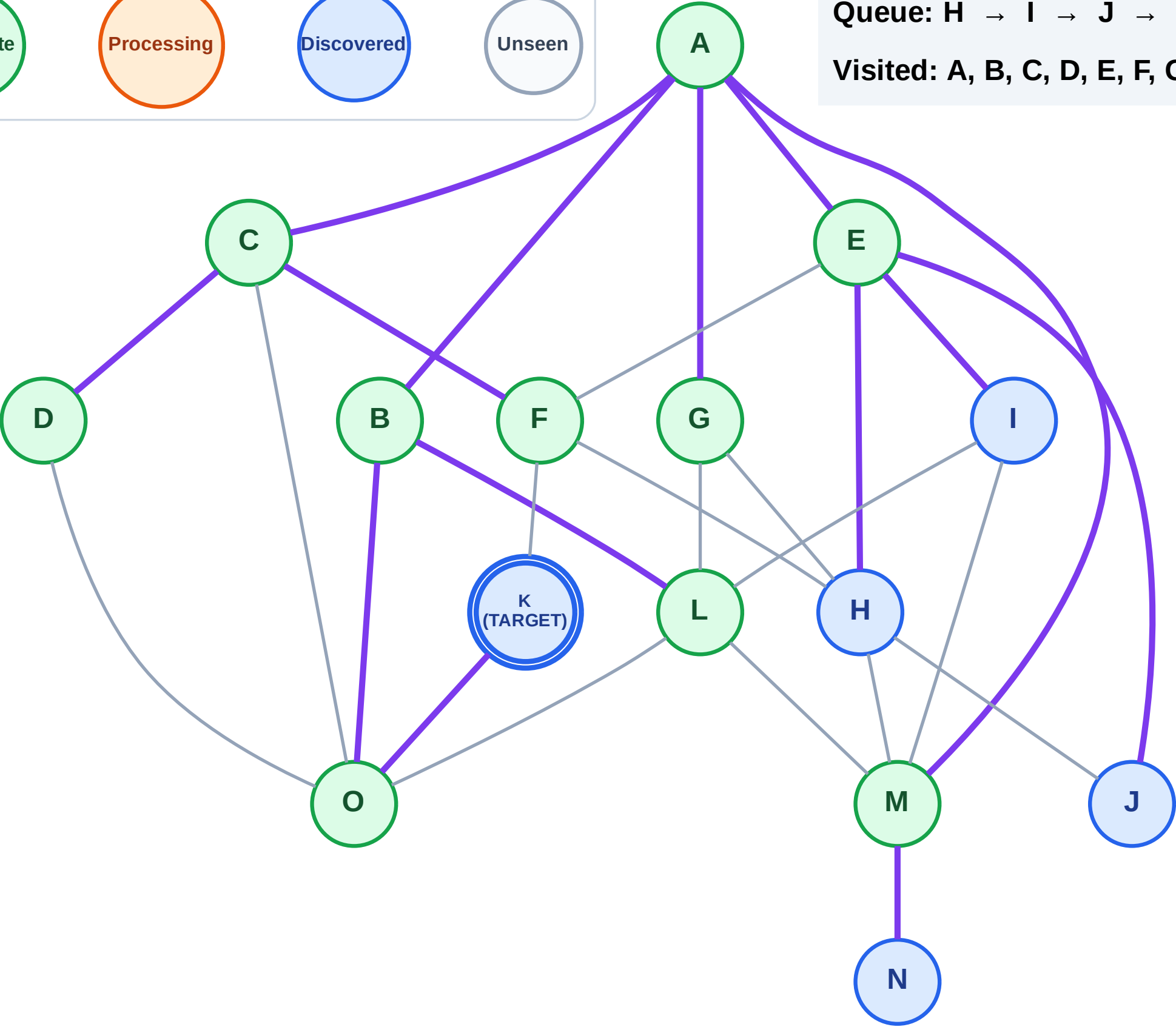
Processing

Discovered

Unseen

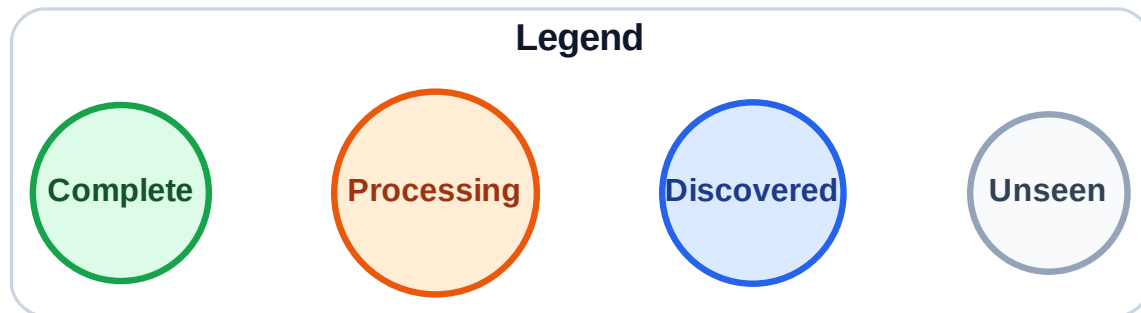
Queue: H → I → J → N → K

Visited: A, B, C, D, E, F, G, L, M, O

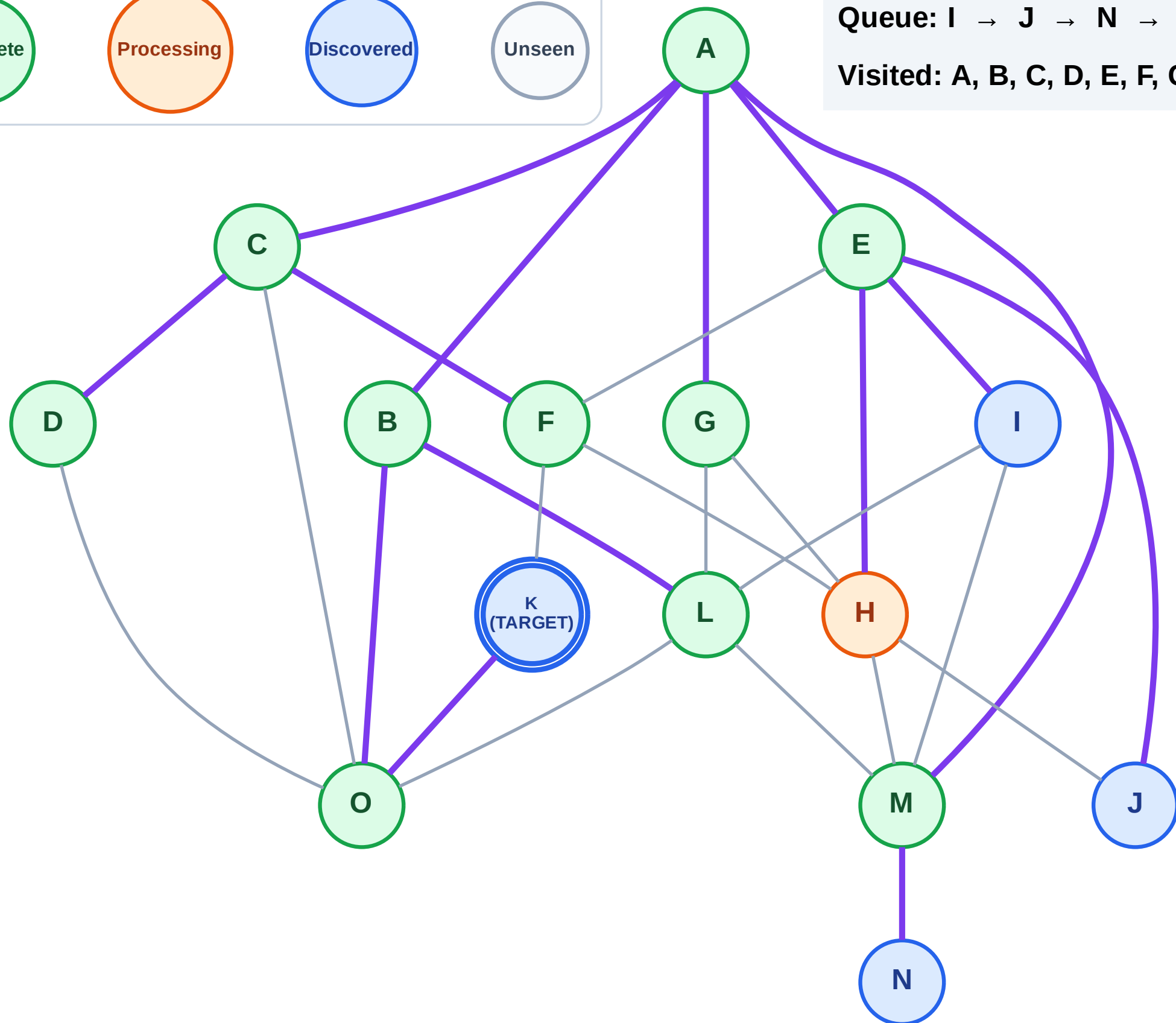


BFS Traversal

Step 22: Process node H



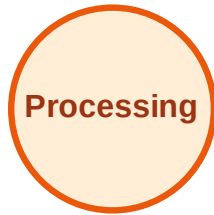
Queue: I → J → N → K
Visited: A, B, C, D, E, F, G, L, M, O



BFS Traversal

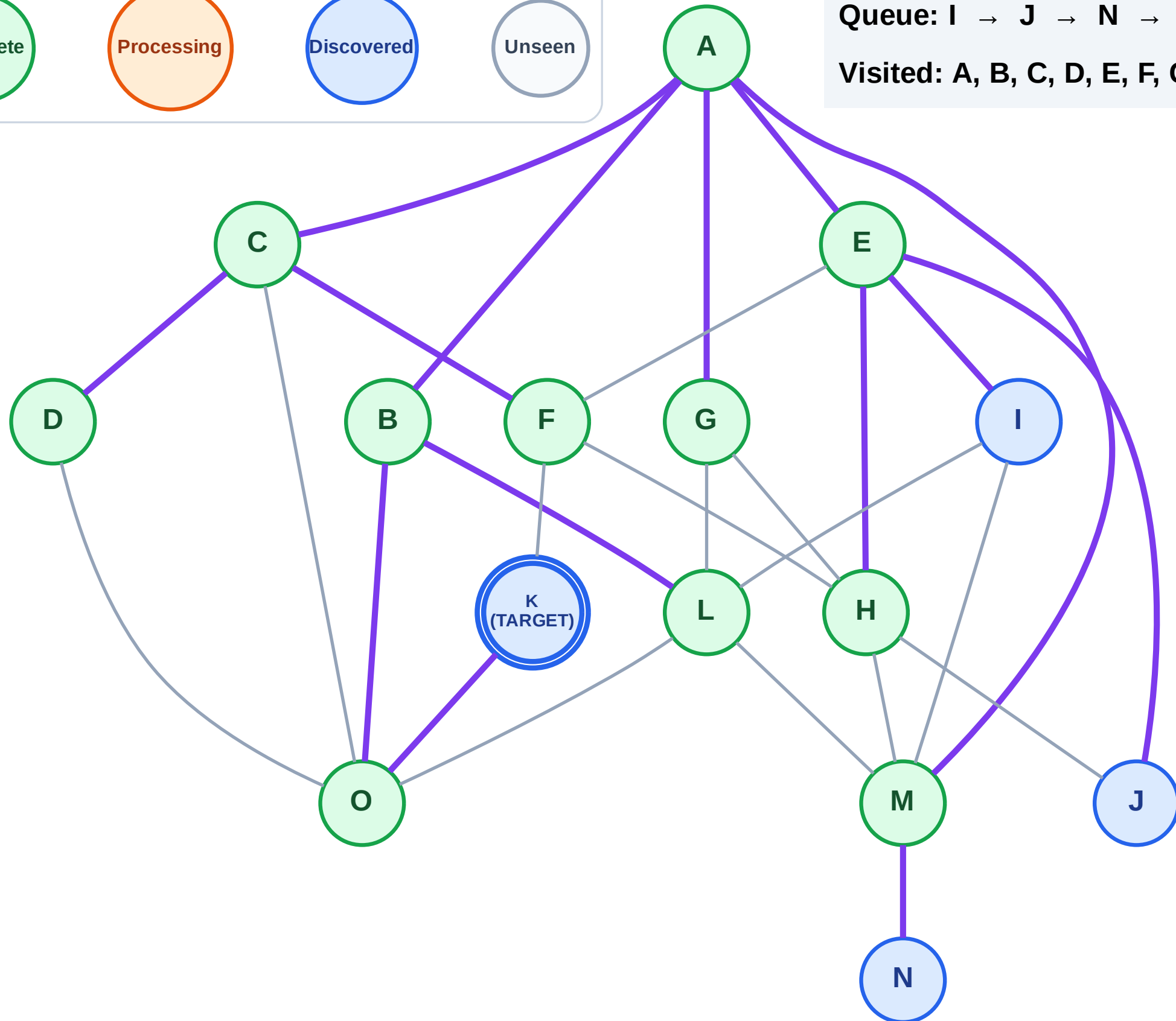
Step 23: No new neighbors from H

Legend



Queue: I → J → N → K

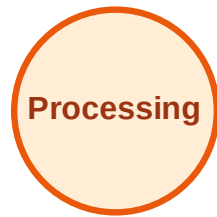
Visited: A, B, C, D, E, F, G, H, L, M, O



BFS Traversal

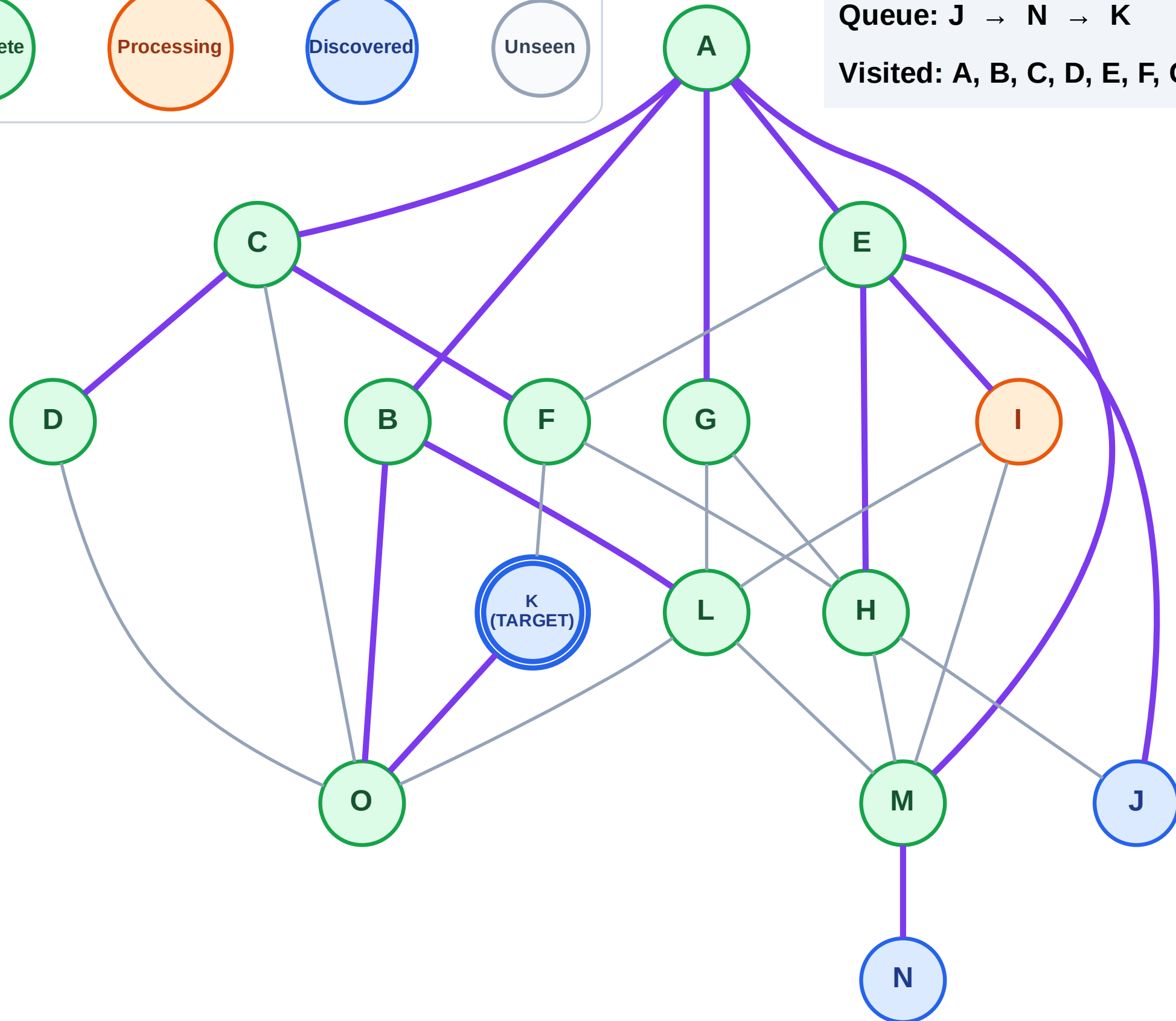
Step 24: Process node I

Legend



Queue: J → N → K

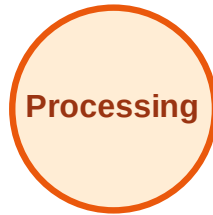
Visited: A, B, C, D, E, F, G, H, L, M, O



BFS Traversal

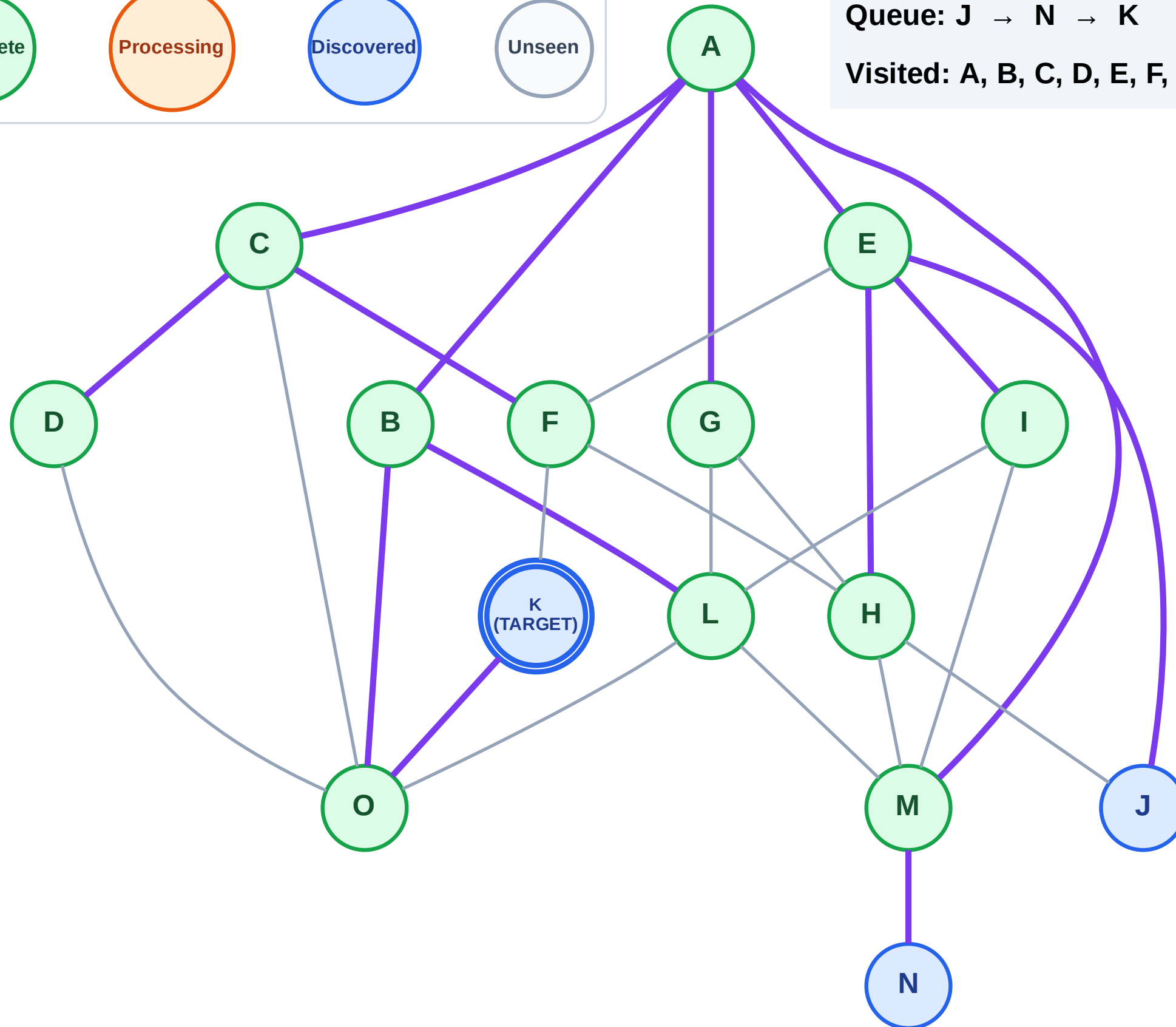
Step 25: No new neighbors from I

Legend



Queue: J → N → K

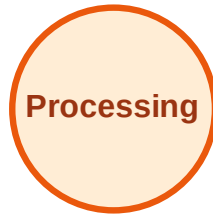
Visited: A, B, C, D, E, F, G, H, I, L, M, O



BFS Traversal

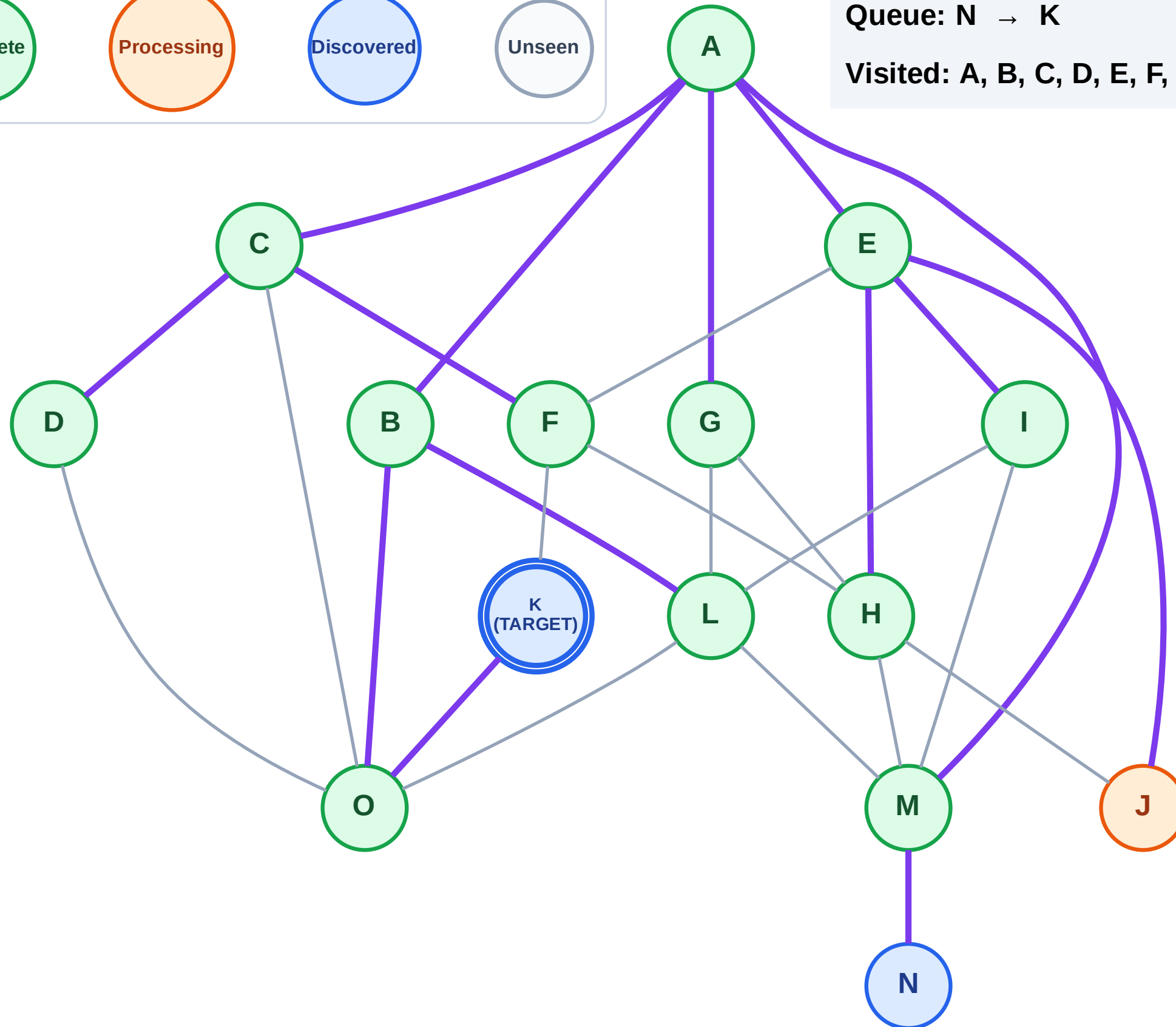
Step 26: Process node J

Legend



Queue: N → K

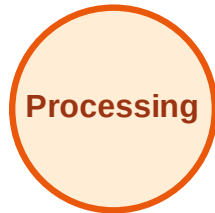
Visited: A, B, C, D, E, F, G, H, I, L, M, O



BFS Traversal

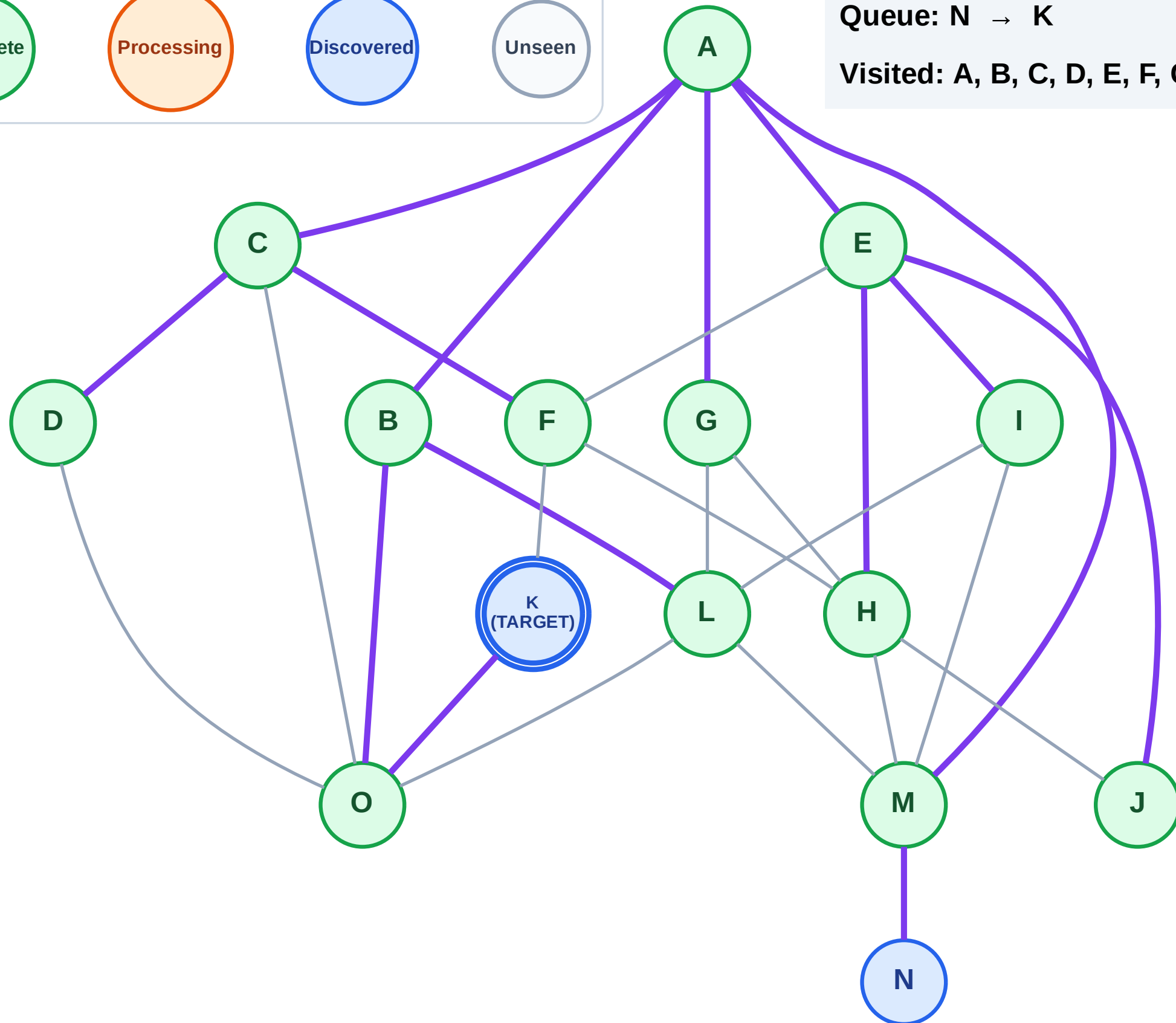
Step 27: No new neighbors from J

Legend



Queue: N → K

Visited: A, B, C, D, E, F, G, H, I, J, L, M, O



BFS Traversal

Step 28: Process node N

Legend

Complete

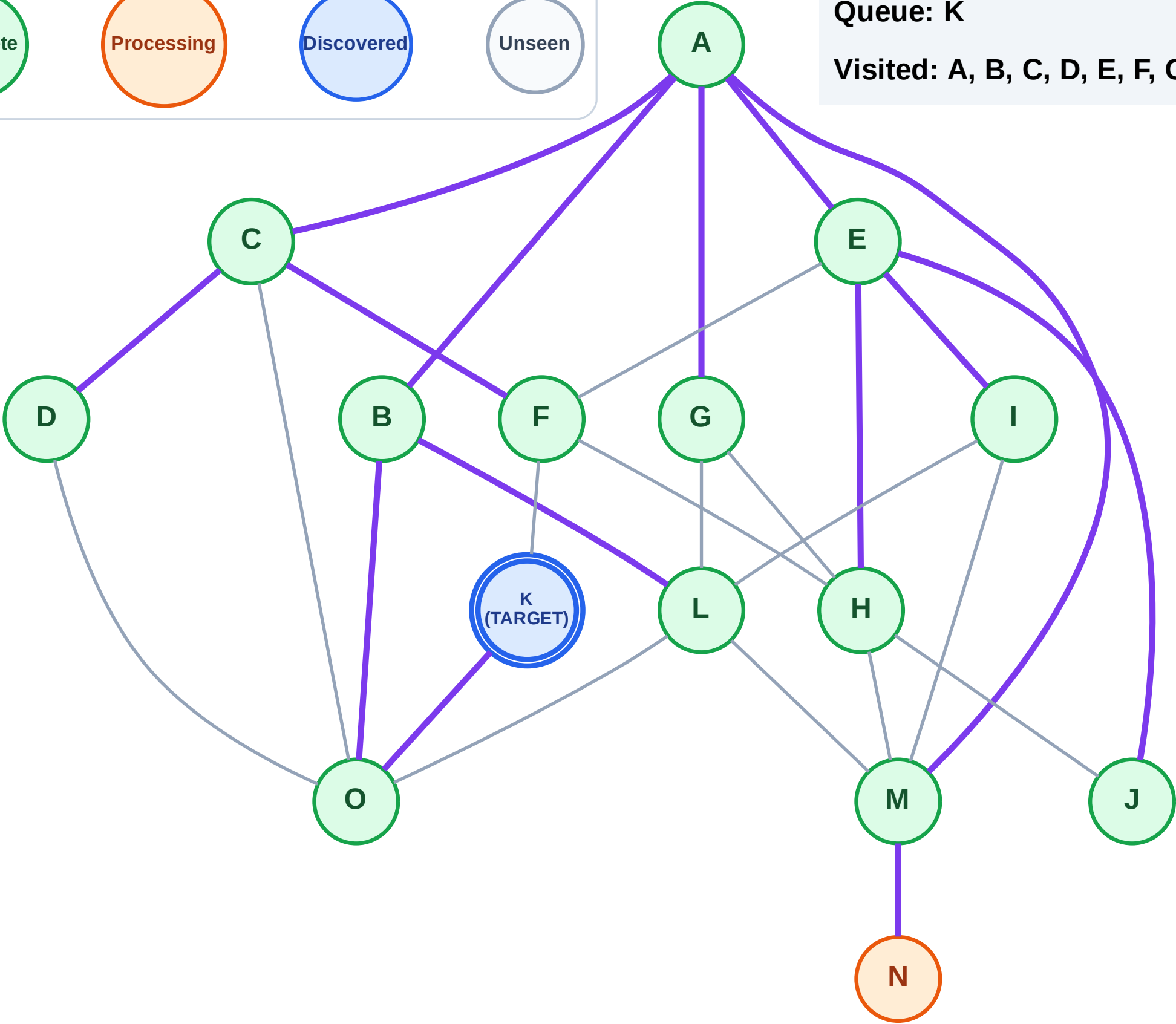
Processing

Discovered

Unseen

Queue: K

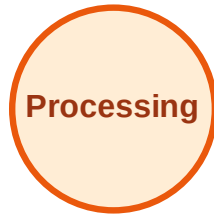
Visited: A, B, C, D, E, F, G, H, I, J, L, M, O



BFS Traversal

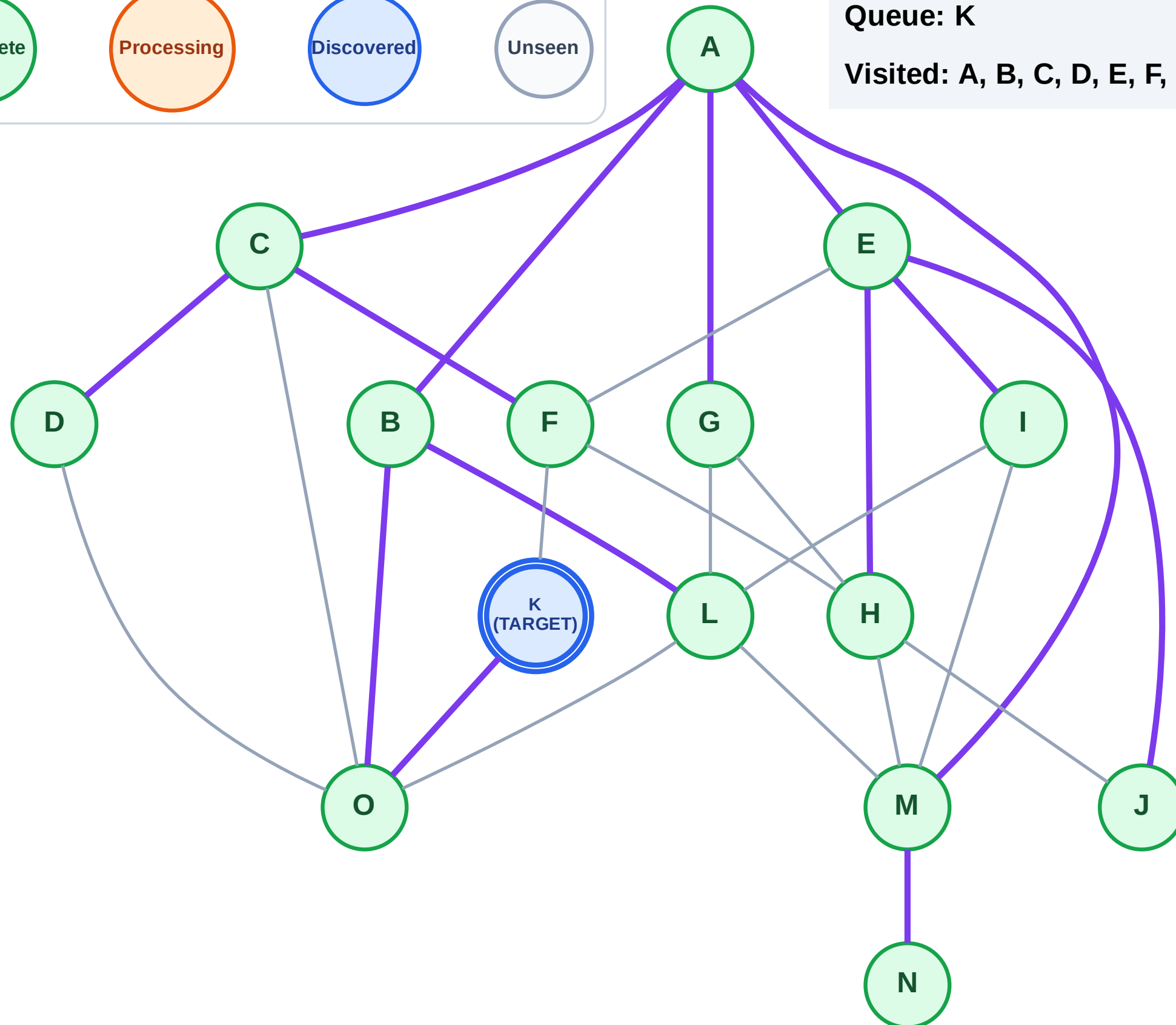
Step 29: No new neighbors from N

Legend



Queue: K

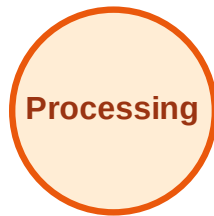
Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



BFS Traversal

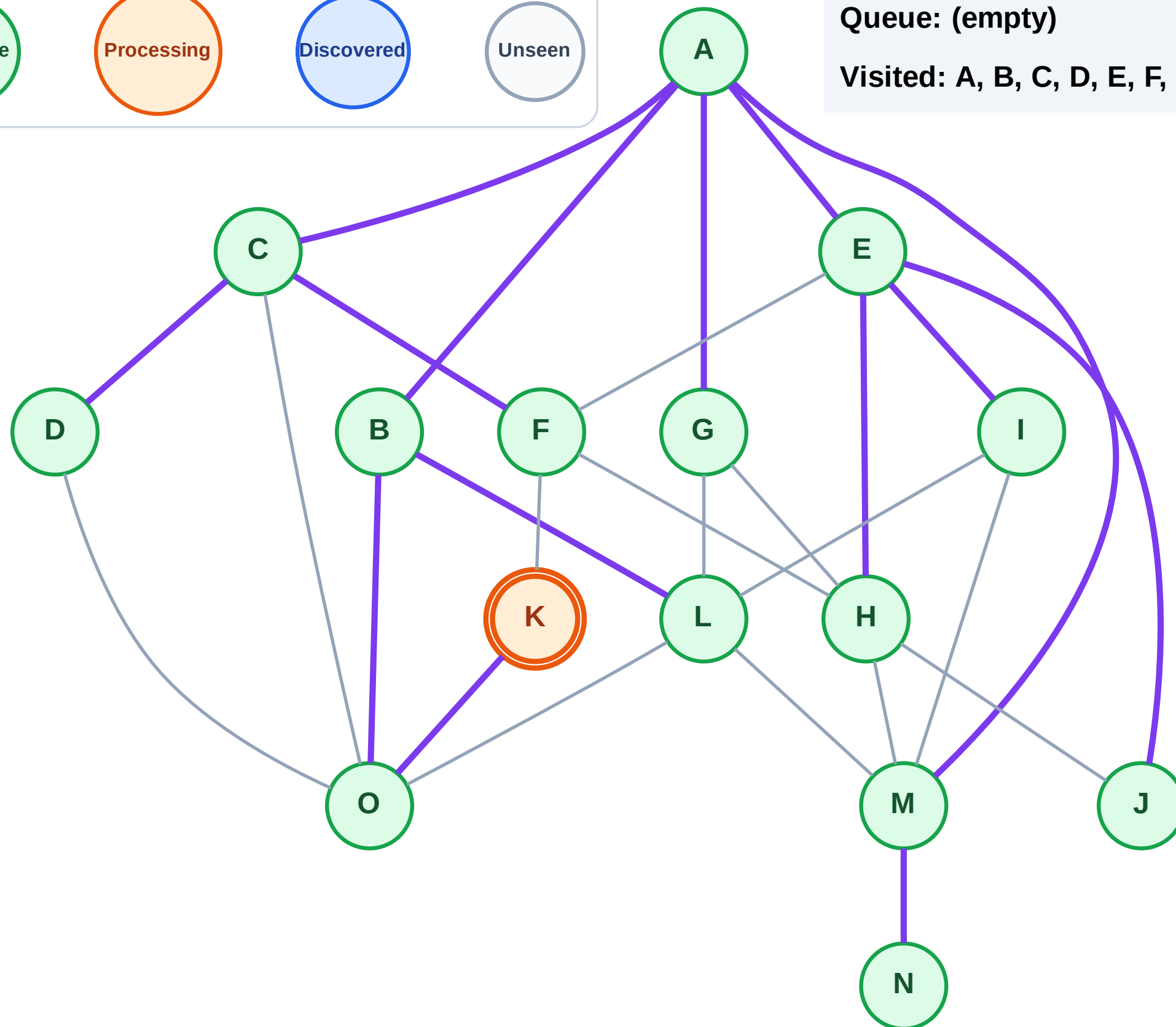
Step 30: Process node K

Legend



Queue: (empty)

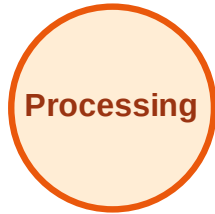
Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



BFS Traversal

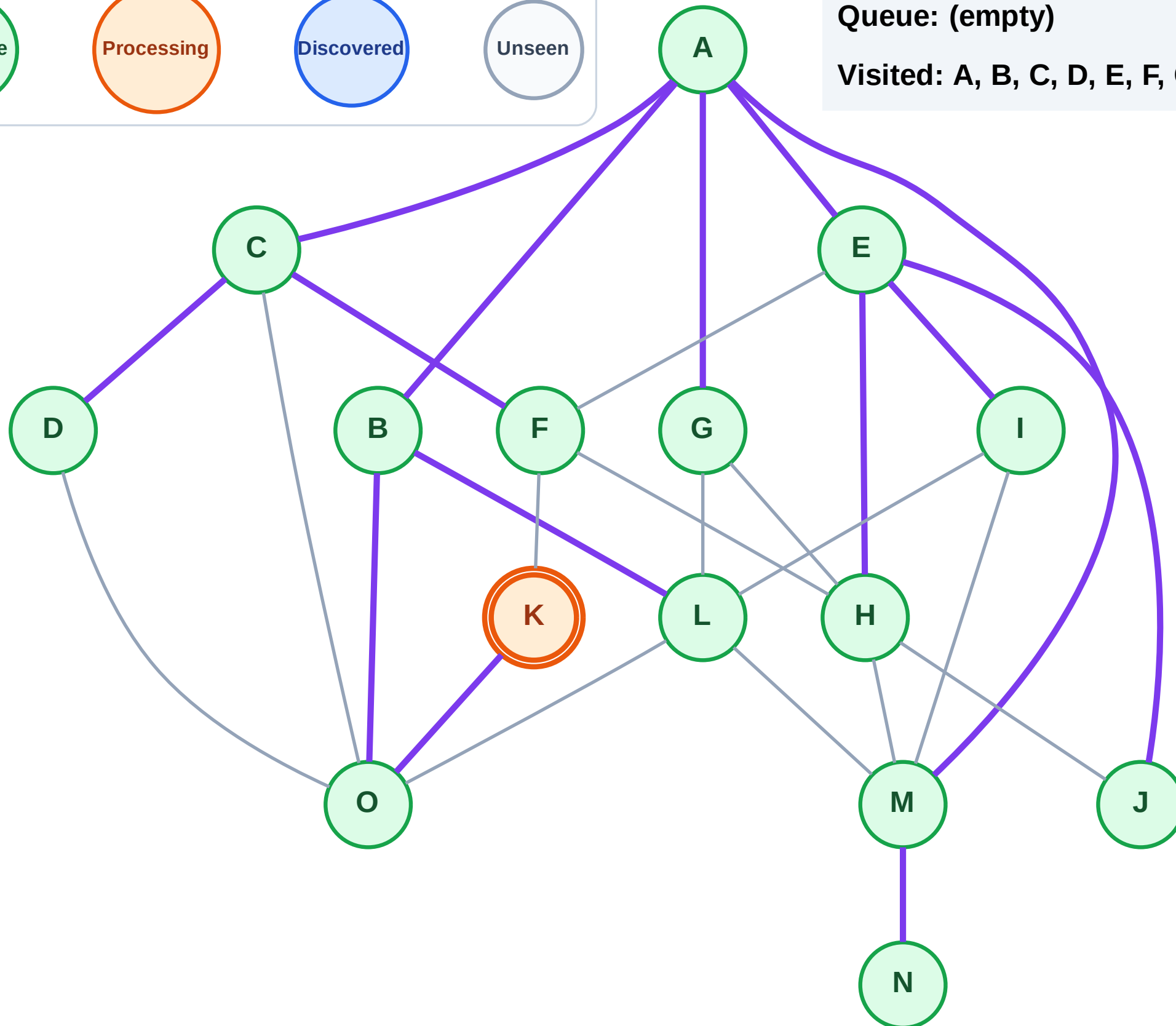
Step 31: TARGET NODE K REACHED! Path (3 hops): A → B → O → K

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

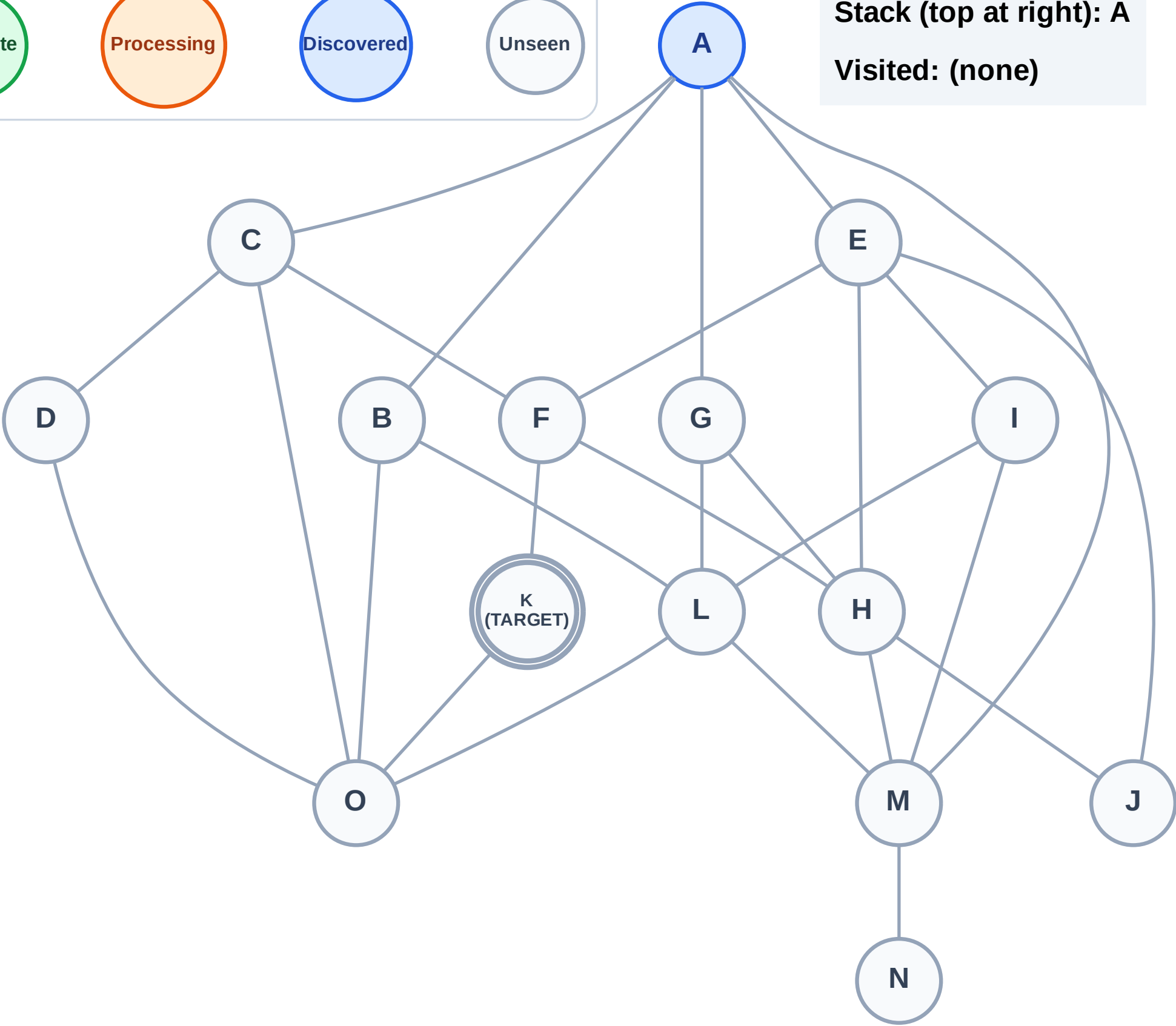
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

Complete

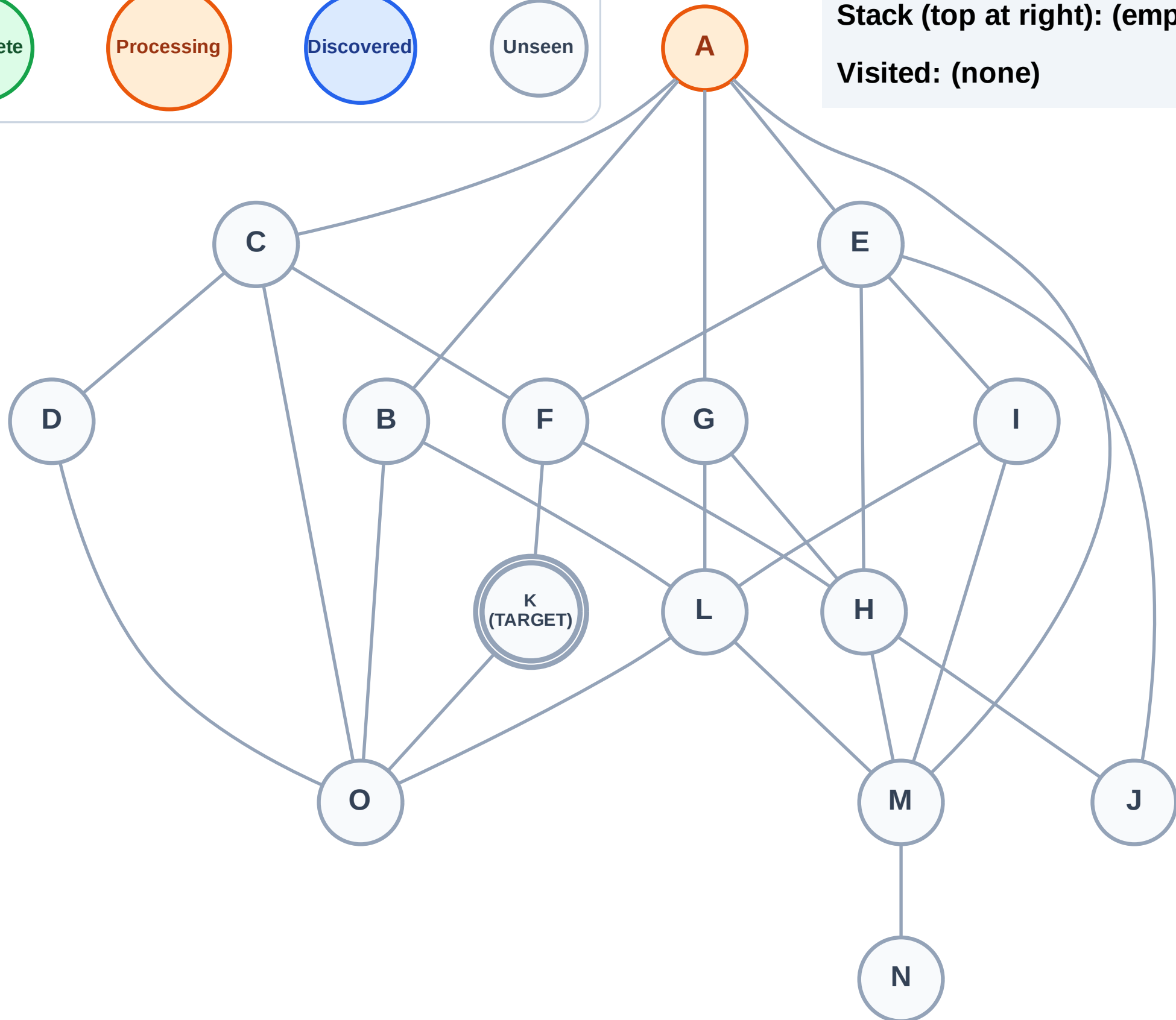
Processing

Discovered

Unseen

Stack (top at right): (empty)

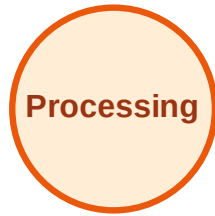
Visited: (none)



DFS Traversal

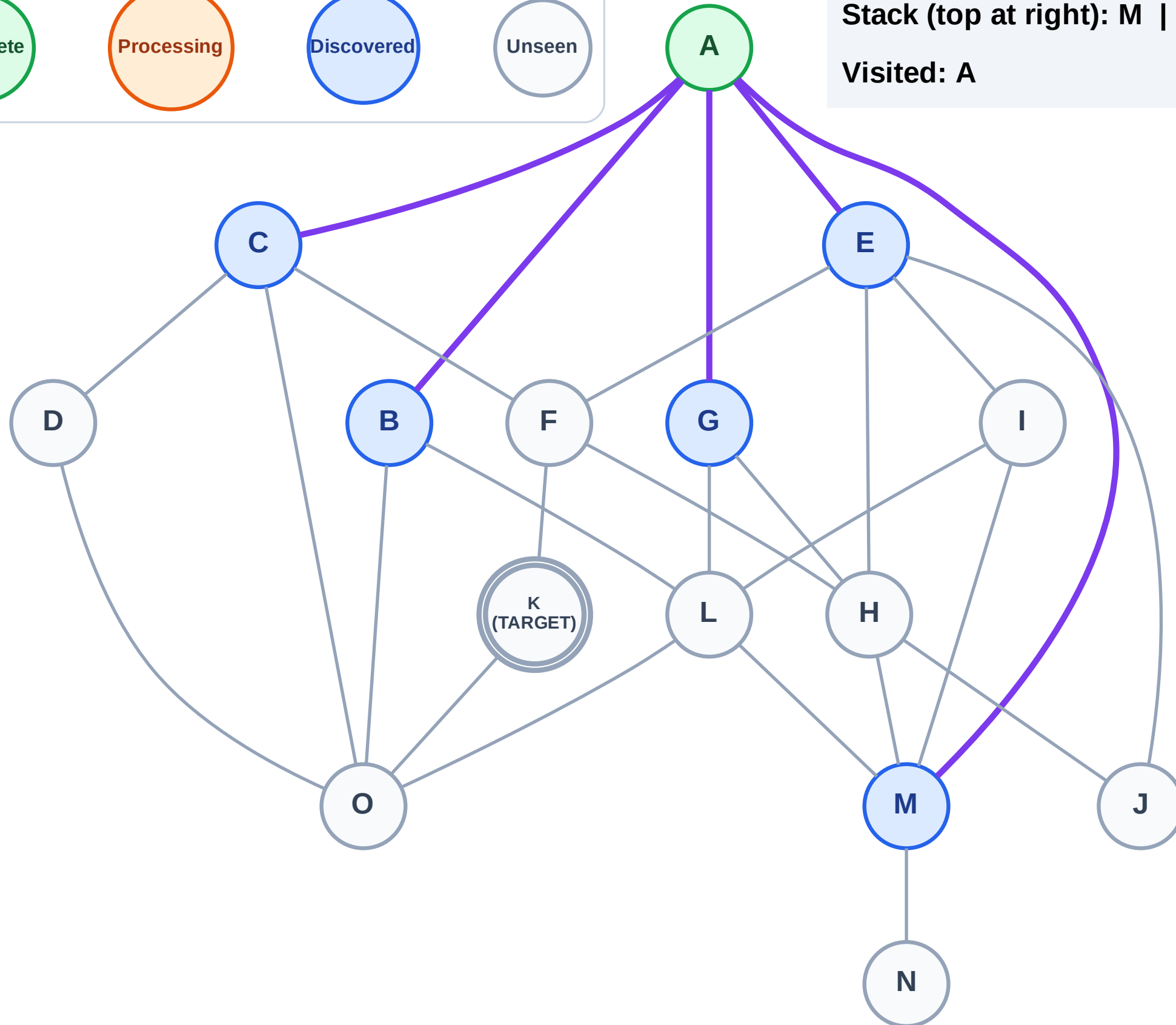
Step 3: Discover M, G, E, C, B from A

Legend



Stack (top at right): M | G | E | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

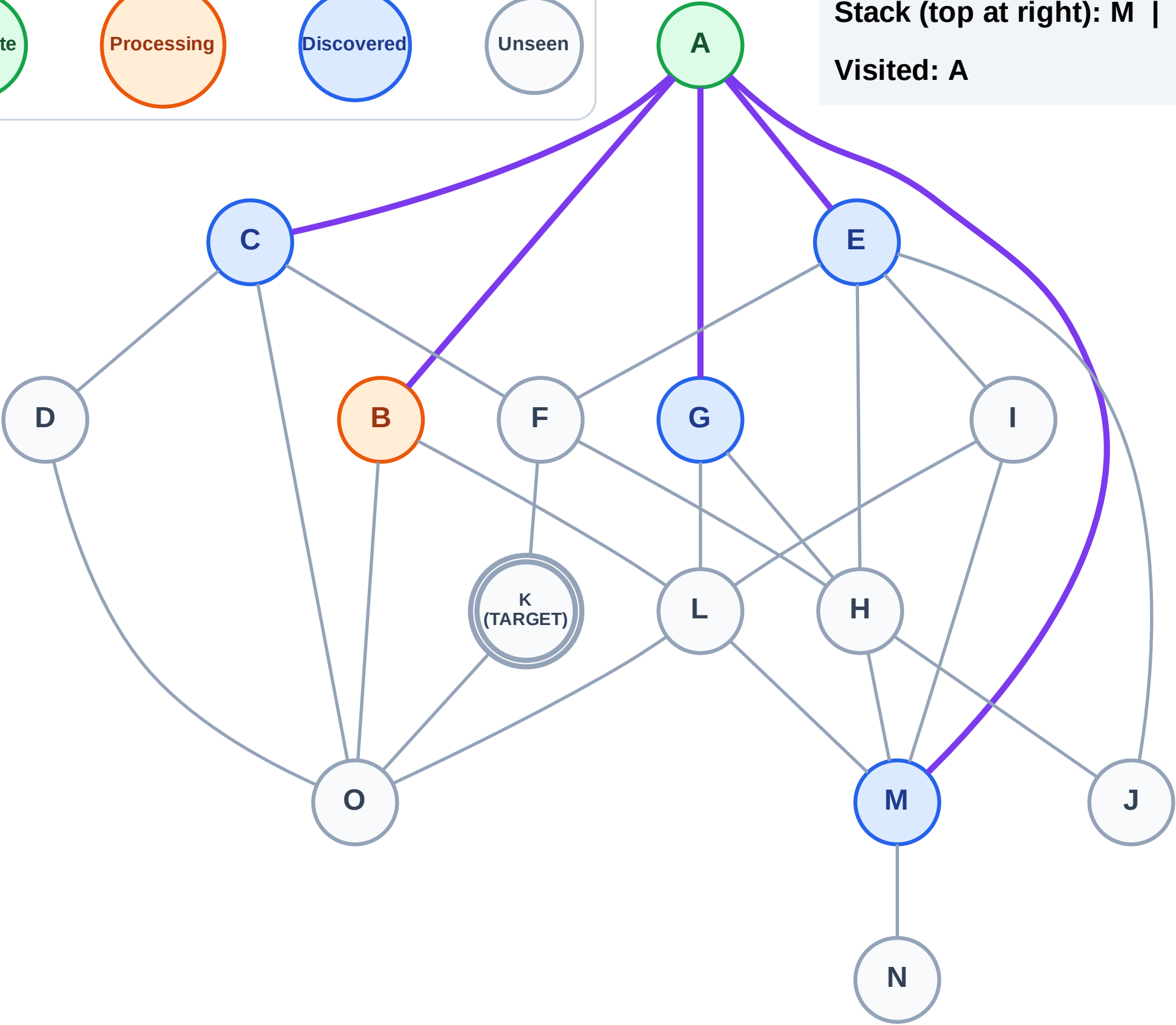
Processing

Discovered

Unseen

Stack (top at right): M | G | E | C

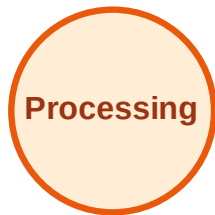
Visited: A



DFS Traversal

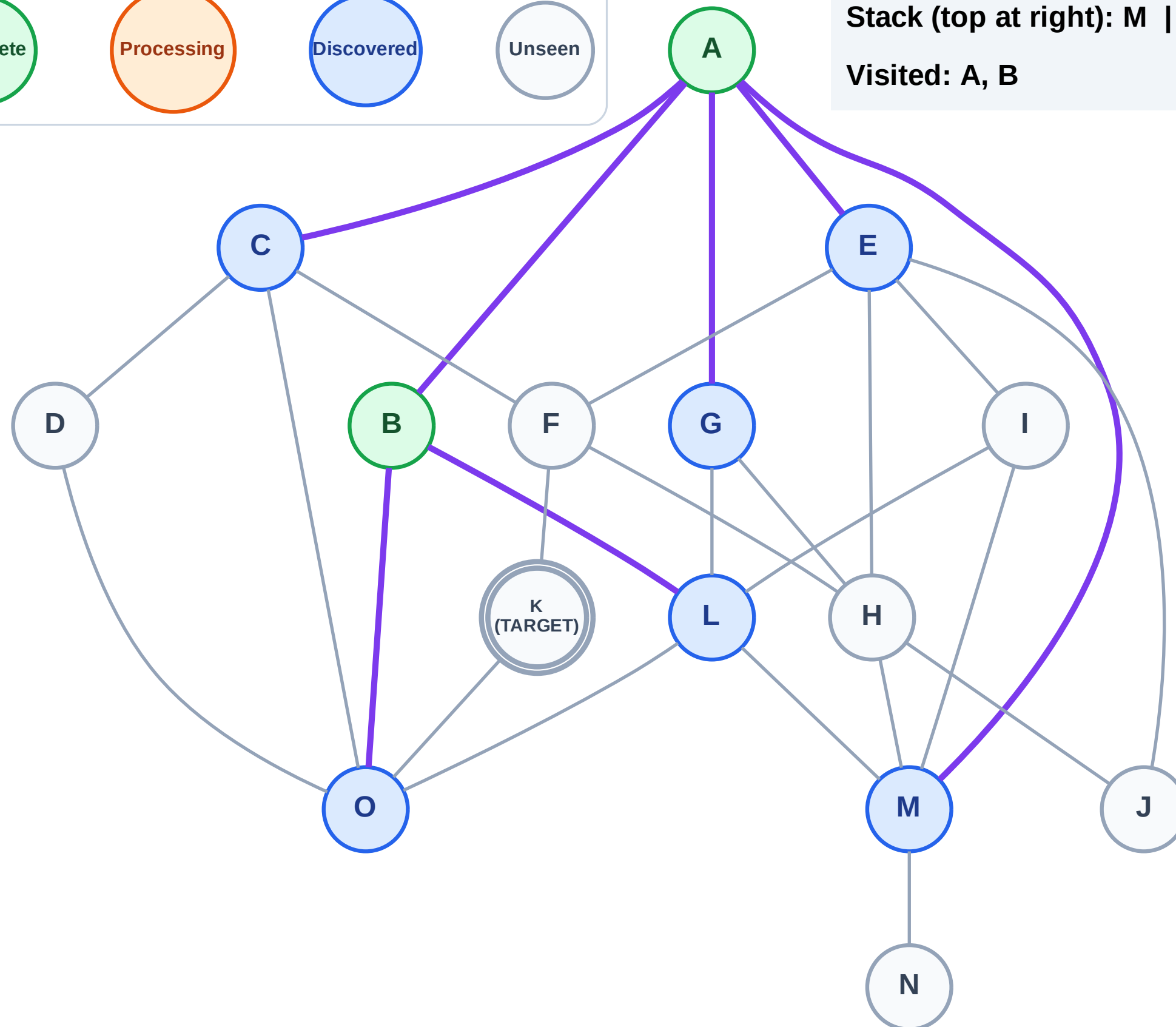
Step 5: Discover O, L from B

Legend



Stack (top at right): M | G | E | C | O | L

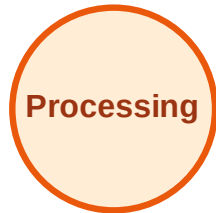
Visited: A, B



DFS Traversal

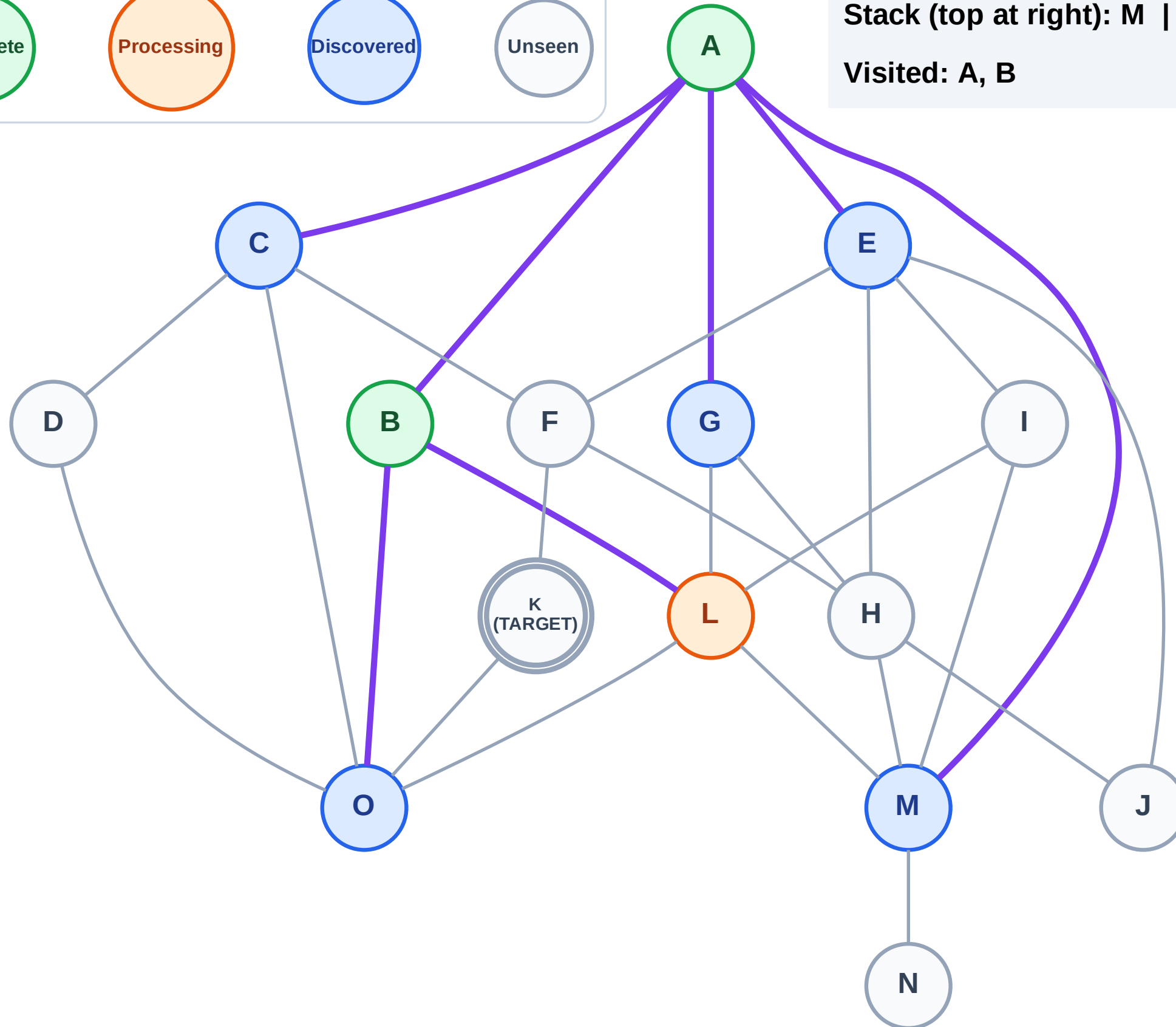
Step 6: Process node L

Legend



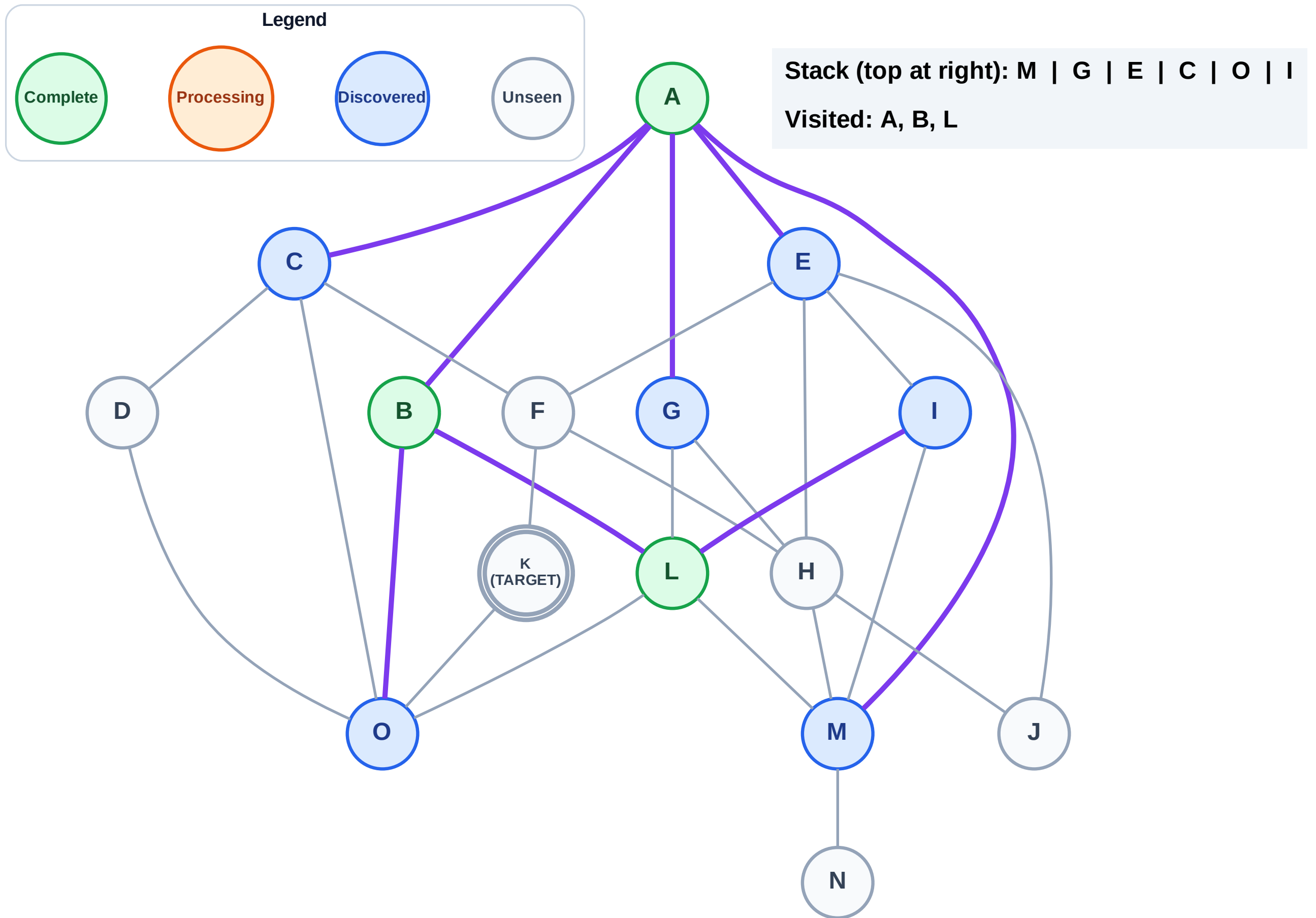
Stack (top at right): M | G | E | C | O

Visited: A, B



Step 7: Discover I from L

Step 7: Discover I from L



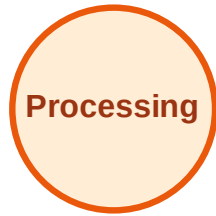
DFS Traversal

Step 8: Process node I

Legend



Complete



Processing



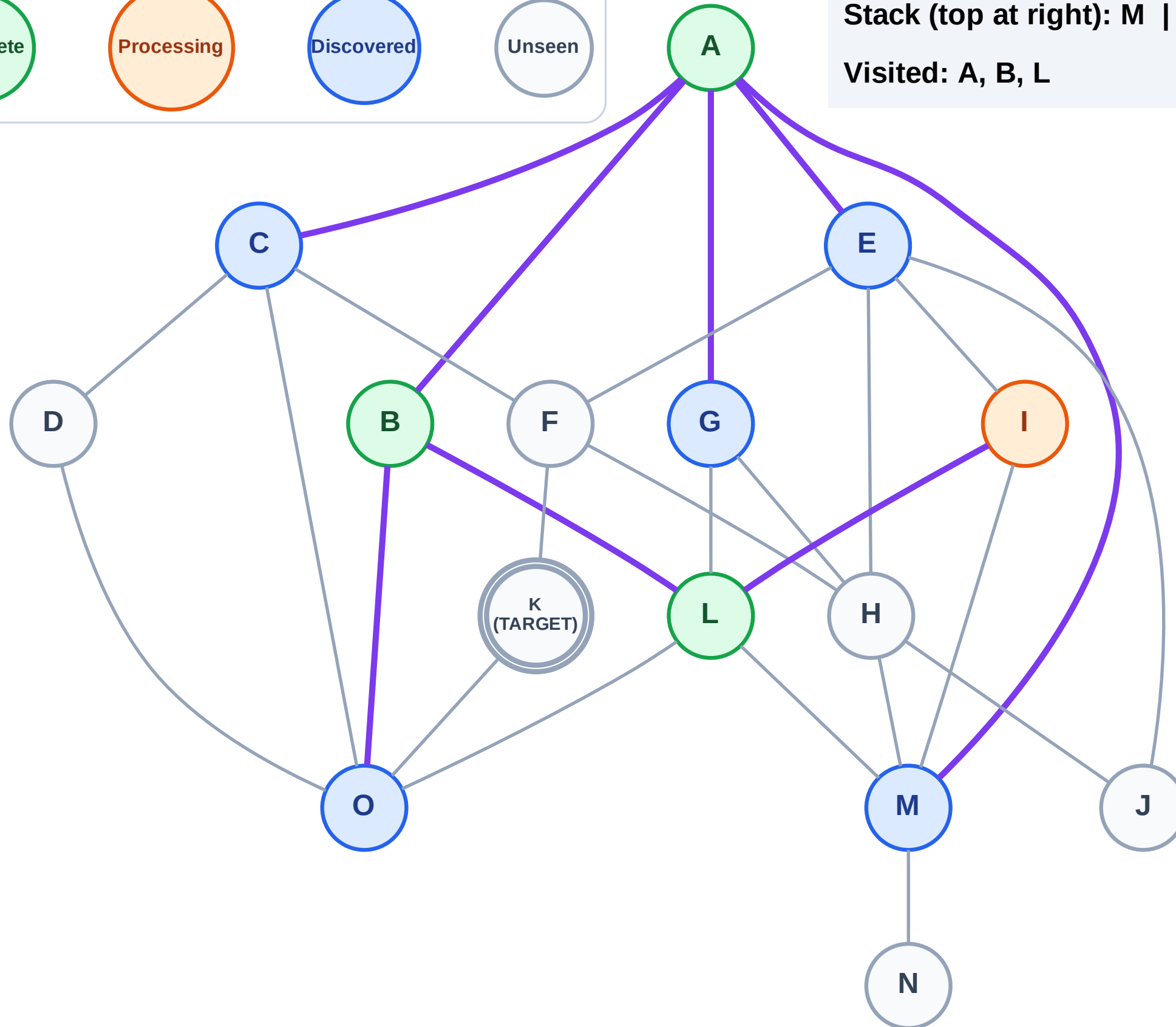
Discovered



Unseen

Stack (top at right): M | G | E | C | O

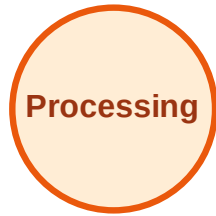
Visited: A, B, L



DFS Traversal

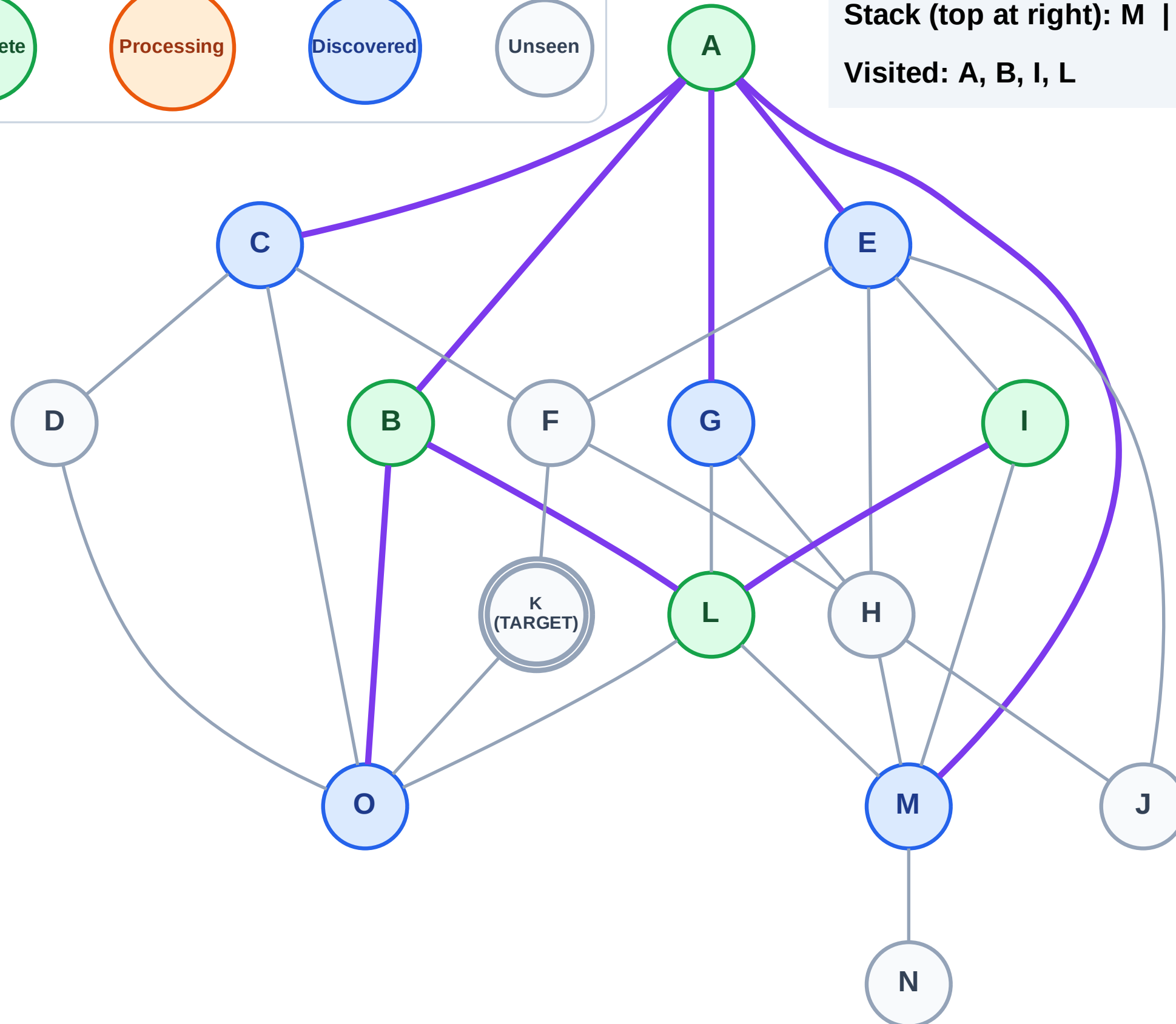
Step 9: No new neighbors from I

Legend



Stack (top at right): M | G | E | C | O

Visited: A, B, I, L



DFS Traversal

Step 10: Process node O

Legend

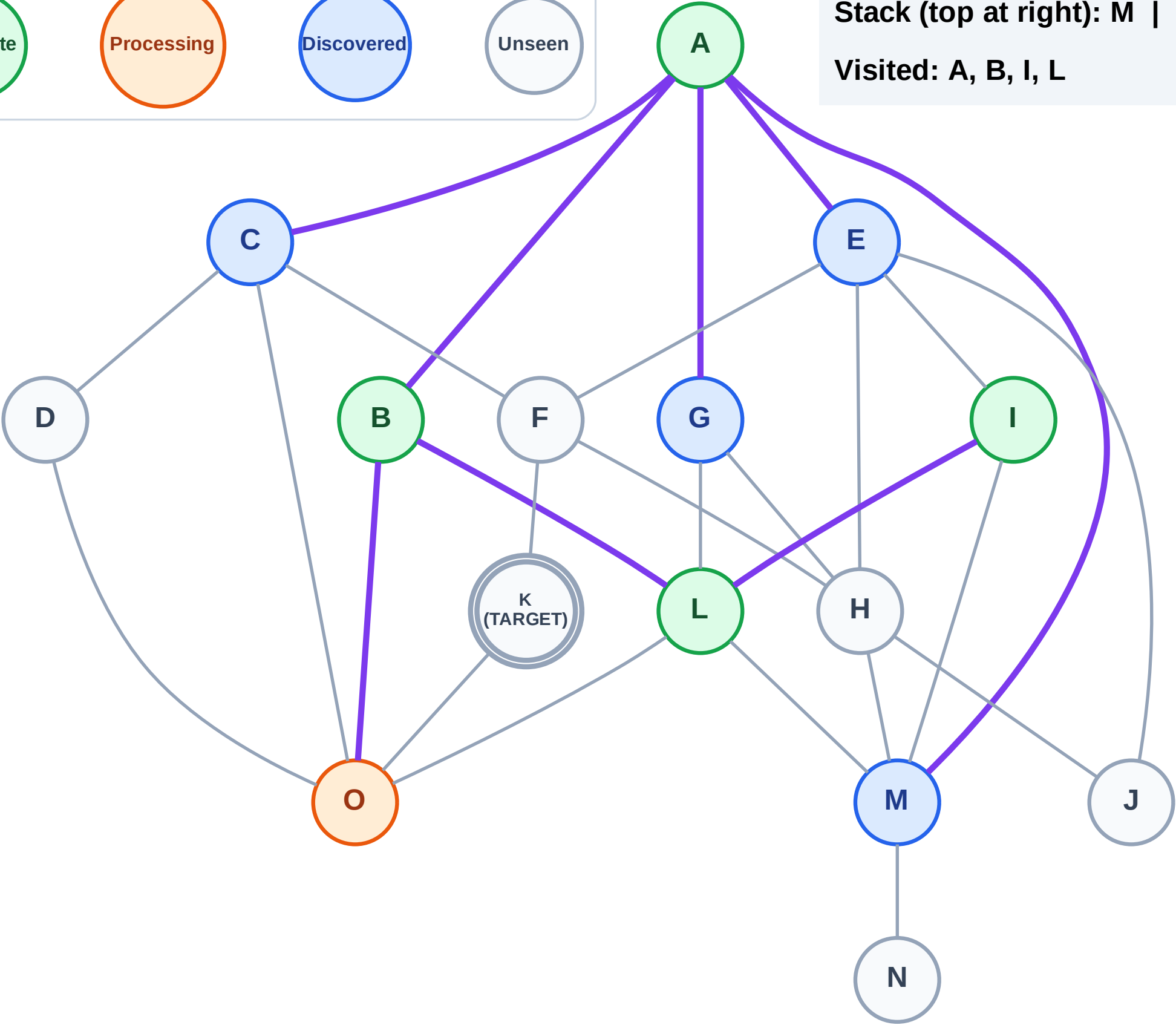
Complete

Processing

Discovered

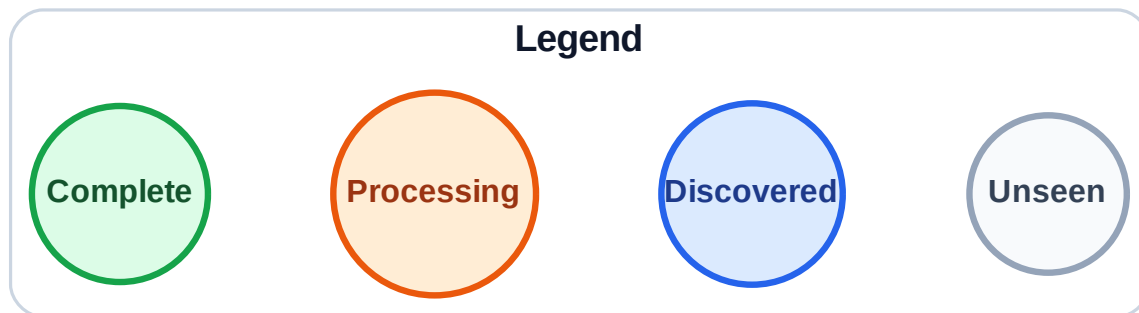
Unseen

Stack (top at right): M | G | E | C
Visited: A, B, I, L



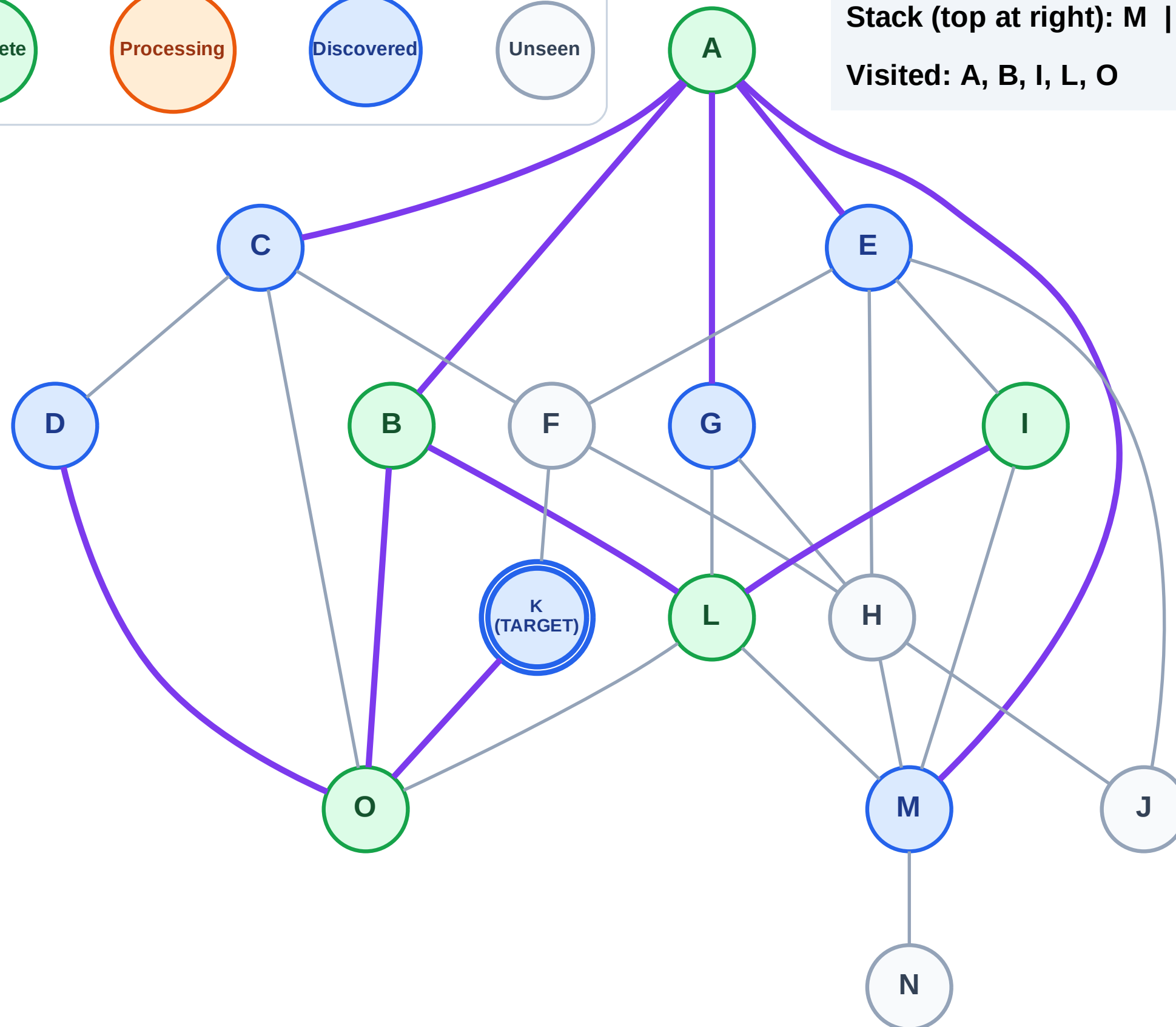
Step 11: Discover K, D from O

Step 11: Discover K, D from O



Stack (top at right): M | G | E | C | K | D

Visited: A, B, I, L, O



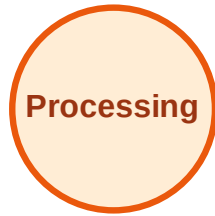
DFS Traversal

Step 12: Process node D

Legend



Complete



Processing



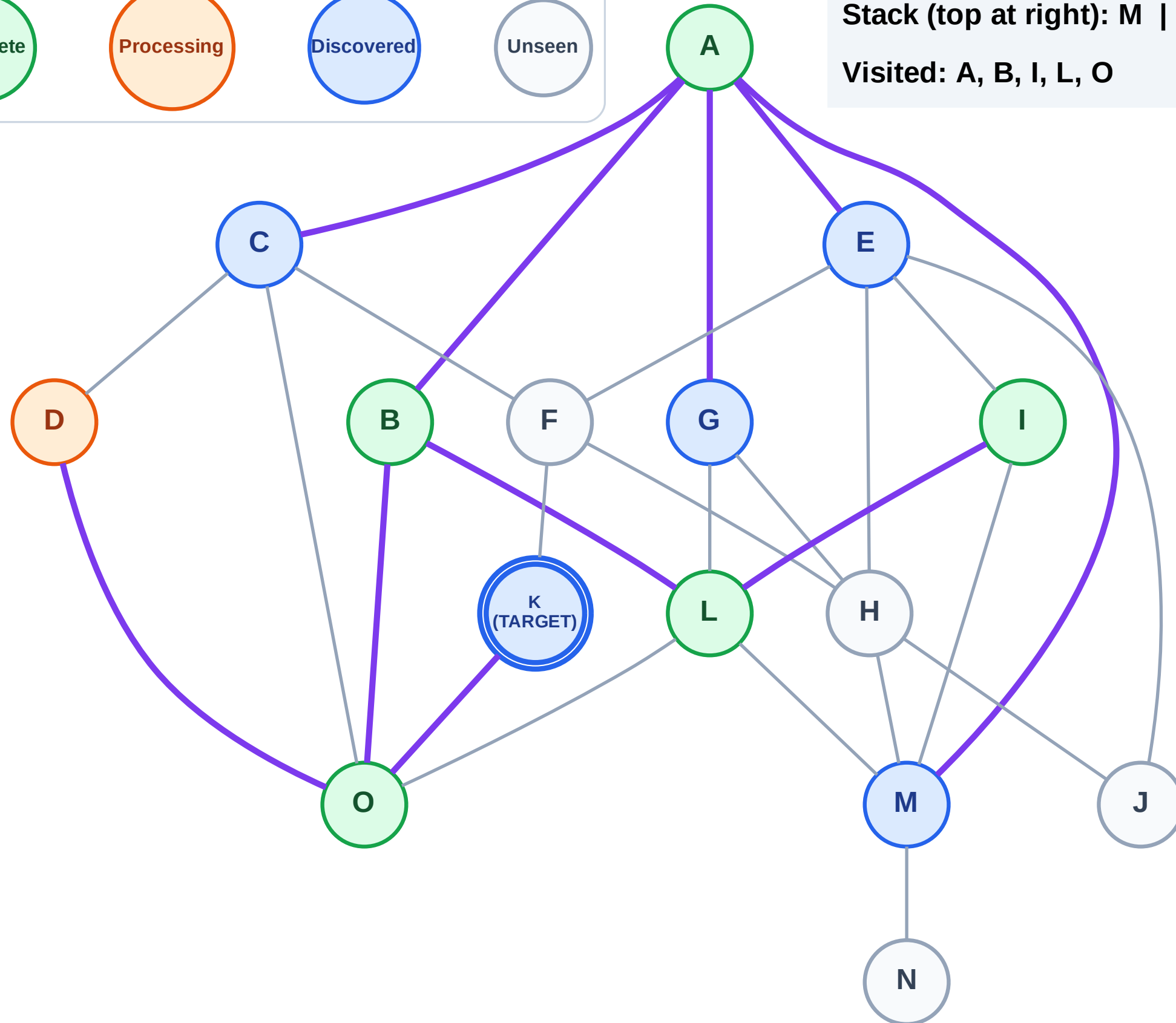
Discovered



Unseen

Stack (top at right): M | G | E | C | K

Visited: A, B, I, L, O

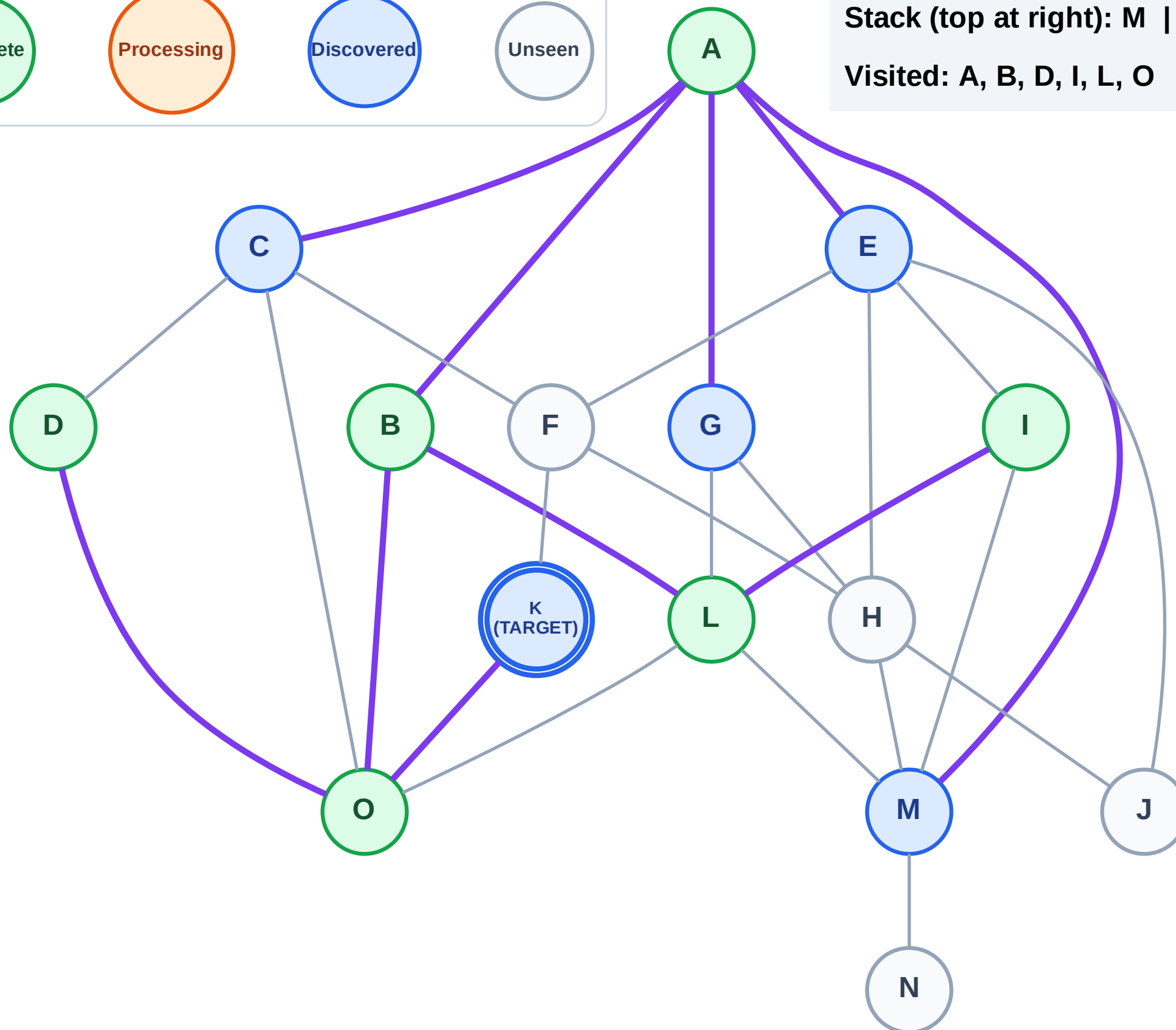


Step 13: No new neighbors from D

Legend

Unseen

Visited: A, B, D, I, L, O



DFS Traversal

Step 14: Process node K

Legend

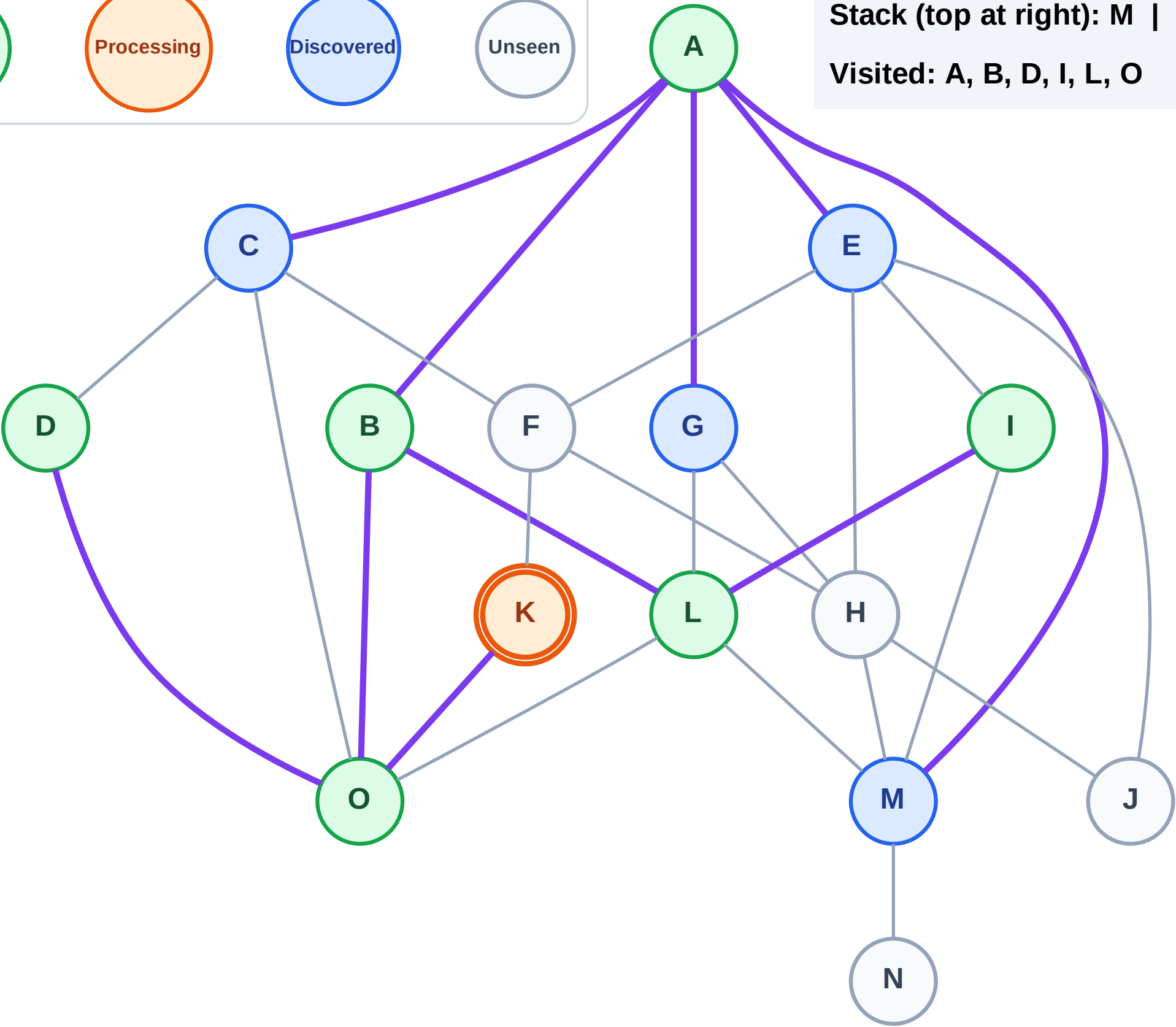
Complete

Processing

Discovered

Unseen

Stack (top at right): M | G | E | C
Visited: A, B, D, I, L, O



DFS Traversal

Step 15: TARGET NODE K REACHED! Path (3 hops): A → B → O → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | G | E | C
Visited: A, B, D, I, K, L, O

